Choupo

The Open Source Chemical Process Simulator

Process Design Heuristics

Choupo-dev

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- the **governing codes and standards** (ASME, API, NFPA, IEC, ISO, EN, and local regulations);
- **detailed design and review by a qualified, licensed professional engineer**;
- a formal **process-hazard analysis** (HAZOP / LOPA) and the **manufacturer's data** for the specific equipment.

Nothing here is a recommendation for a real plant, and no value has been validated for any specific service. **Safety-critical decisions** — pressure relief, fired equipment, toxic or flammable inventories, electrical area classification, effluent limits — *can cost lives or breach the law if wrong* and MUST follow the applicable standard and competent review, never a rule of thumb. Where a number could touch safety or the environment, the controlling standard is named (see *Standards & further reading*).

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This guide is Choupo's answer to the rules-of-thumb appendices you reach for in Turton, Seider, and Sinnott & Towler — but rebuilt for a simulator whose whole point is that *nothing is hidden*. Where a textbook table hands you a single number ("use ΔT_min = 10 °C", "size liquid lines at 1–3 m/s"), this guide hands you a **range, a *why*, and a runnable hypothesis**. The eight faculty sections that follow cover the full sweep of decisions a student faces when a problem stops being "solve this flash" and becomes "what equipment do I even pick?": process structure, reactor type, separation train, exchanger type and velocity, line sizing and ΔP budgets, membranes, solids handling, and the metal envelope that contains all of it.

How to read every rule here: a heuristic is a *hypothesis*, not a verdict. "Pick a PFR when you need high conversion of a single reaction" (the reactor rule below) is a starting bet that buys you 80 % of the answer in 20 % of the effort — it is *not* a substitute for the run. Choupo is glass-box precisely so you can test the bet: build the case, watch the Newton iterations, read the KPIs, and confirm (or refute) the rule for *your* feed, *your* thermodynamics, *your* pressure level. Wherever a rule maps cleanly onto an existing tutorial, the section points you at it — run it, perturb it, and see the range move. A heuristic you have personally falsified once is worth ten you merely memorised.

Why the sections are ordered the way they are — the design onion. Process design proceeds from the inside out: the reactor sits at the core, the separation-and-recycle system wraps it, the heat-recovery network wraps that, and the utilities, hydraulics, and mechanical envelope wrap the whole. Each layer's decisions *constrain* the next — the reactor's conversion fixes what the separation train must handle; the separation train's streams fix the heating and cooling duties; those duties fix the pumping power and line sizes; and the worst temperature, pressure, and corrosivity anywhere inside fix the materials of construction on the outside. So we read **outermost framing first** (synthesis), then **inward to the core** (reactors), then **back out through the layers**: the full separation train kept together (distillation → membranes → solids), then heat recovery (heat exchangers), then the connective hydraulics, and finally the material envelope that contains everything. This keeps every decision in the order in which its inputs actually become available.

1 Process synthesis & flowsheet structure

Design the flowsheet **inside-out**, following Douglas' onion. Each layer fixes the degrees of freedom (DOF) of the next, so never optimise an inner layer against an outer-layer guess.

Onion layer	What you DECIDE	Heuristic numbers / ranges	Physical/economic WHY
1. Reactor	CSTR vs PFR, single-pass conversion X	Push X to **0.6–0.9** for cheap reactant; stop where recycle cost rises faster than reactor cost. Keep $\mathbf{X \leq 0.95}$ if selectivity falls (consecutive rxns)	Reactor cost $\propto$ V $\propto$ ln[1/(1−X)] for a first-order PFR; recycle/separation cost $\propto$ unconverted feed. The minimum-TAC X sits where ∂(reactor)/∂X = −∂(recycle)/∂X
2. Separation	Recover vs recycle; sequence	Recover species worth > ~**3–5**× its separation cost; aim **99%+** recovery of valuable product	Lost product is pure margin; over-purifying a recycle wastes reboiler duty
3. Recycle/purge	Purge fraction for inerts	Purge so inert mole-frac in loop ≤ **0.10–0.30**; purge loss of reactant < **1–5%** of feed	Inert build-up raises recycle compressor/pump duty ($\propto$ total recycle flow); too-small purge → runaway loop inventory
4. Heat recovery	HEN, ΔT_min	ΔT_min **10 °C** liquid–liquid, **20–40 °C** with a gas stream, **3–5 °C** cryogenic/refrigerated	Area $\propto$ 1/ΔT_min (U_gas ~15–60 W/m^{2}K vs U_liq ~300–1000); small ΔT buys energy but explodes area cost
5. Utilities	Hot/cold service, fired vs salt	LP/MP/HP steam below ~250 °C; **Dowtherm A** to ~400 °C; **HiTec salt 142–535 °C**; fired heater only above salt range	Match utility T just above pinch; over-hot utility destroys exergy and over-sizes ΔT

Input–output structure first. Before any unit, draw the I/O box: feeds in, products + purge + by-products out. Set **feed stoichiometric ratio** (slight excess, ~1.05–1.2, of the cheaper/limiting-safety reactant), decide which species **leave vs recycle**, and locate the **purge** (needed whenever a feed impurity or light by-product has no other exit). This fixes overall material balance DOF *before* you size anything.

Reactor type. Use a **PFR** when you need high conversion of a single reaction (plug flow → smaller volume for given X) or want a temperature/composition profile; use a **CSTR** for liquid-phase, exothermic, fast, or solids/biological systems needing isothermality and mixing. For series-parallel selectivity, a **CSTR-then-PFR** or staged CSTRs (3–4 stages approach PFR) is common. See `tutorials/steady/reactors` cases; for economics-driven X, `optim05_reactor_npv`.

Distillation sequencing (Lightest-out / direct sequence is the default): (1) remove **most plentiful** component early; (2) take the **lightest** (highest relative volatility, α) split first when α > ~1.5; (3) leave the **hardest split** ($\alpha \to 1$) and the **highest-purity** product for last, alone, to avoid re-contaminating it; (4) favour **near-50/50** splits to balance column loads. Each ordering is a hypothesis — run the Wang-Henke / FUG shortcut cases (`tutorials/steady/distillation`) and compare reboiler+condenser duty.

Degrees of freedom discipline. A recycle adds one tear stream; Choupo's Newton-on-tears solver converges it, but **you** still own the design specs (purge fraction, reactor X, reflux). Count them: overall DOF = stream variables − independent balances − fixed specs. If the flowsheet won't close, you've under- or over-specified — the glass-box log tells you which. Sequence the whole sheet **reactor** → **separation** → **recycle** → **HEN** → **utilities**, re-pinching only after the recycle converges, because recycle flows change the composite curves.

2 Reactors & reaction engineering

The reactor is rarely the most expensive vessel, but it sets the composition every downstream unit must clean up — so its selection ripples through the whole flowsheet cost. Pick the **type** from mixing/kinetics, then size from **residence time** and **heat load**.

2.1 Type selection

Criterion	Batch / semi-batch	CSTR	PFR / tubular	Fixed bed (catalytic)	Fluidised bed
Throughput	< 500 t/yr, multiproduct	medium, liquid	medium–high	high, gas+solid cat.	high, fast-deactivating cat.
Phase	liquid, slurry	liquid, L–L, gas–liq	gas or liquid	gas/liquid + solid	gas + fine solid
Kinetics favoured	slow (hrs)	any; needs CSTR robustness	high order, fast	heterogeneous	very exo/endothermic, ΔT-sensitive
Conversion per pass	high (time-driven)	**lowest for given V**	high	high	moderate
Heat control	jacket/coil, batchwise	excellent (isothermal, well-mixed)	hard (axial profile)	hot-spots; multitube	near-isothermal (solids mix)
τ for same X	—	**largest**	smallest	small	small

The governing trade-off: a CSTR operates at the **outlet** (low-reactant) concentration, so for n>0 kinetics it needs the largest volume — typically **1.5–4×** a PFR for the same conversion (Choupo `cstr01_first_order` vs `pfr01_first_order` shows this head-to-head). You accept that penalty to buy near-perfect temperature control and the ability to run slurries/fouling systems a tube would plug. Use **CSTRs-in-series** (3–5 tanks) to approach plug-flow behaviour while keeping the mixing robustness.

2.2 How far to push conversion

Per-pass conversion is an **optimisation, not a maximisation**. Reactor volume scales roughly as `V` $\propto$ `ln[1/(1-X)]` (first order) — the last few percent of conversion cost disproportionate volume and, worse, residence time that breeds side products and erodes selectivity. **Heuristics:**

- Single-pass irreversible, valuable feed: target **X = 0.6–0.9**, then **recycle unconverted reactant** (recovery + recycle is almost always cheaper than the extra reactor volume; see `conversion01_hda` and the recycle flowsheet `process01_reactor_flash`).
- Reversible reactions: stop at **80–95% of equilibrium X**; chasing equilibrium needs infinite τ (`cstr02_reversible_wgs`, `pfr02_reversible_wgs`).
- The economic optimum trades reactor capital vs separation/recycle duty — run a sweep; `optim05_reactor_npv` does exactly this on NPV.

2.3 Temperature, pressure & exotherm policy

Run **as hot as selectivity and catalyst/material limits allow** — rate roughly doubles per **+10 °C** — but watch the Arrhenius gap: if the side reaction has higher E_a, hotter means *less* selective. For reversible exothermic reactions use a **declining T profile** (hot inlet for rate, cool outlet for equilibrium).

Heat removal sets the geometry. Estimate **adiabatic ΔT_ad = ($-\Delta$H_rxn) $\cdot$ X $\cdot$ C_feed / ($\rho \cdot$ cp)**; if ΔT_ad exceeds ~**20–50 °C**, an adiabatic vessel is unsafe — go to cooled multitube, CSTR with coil/jacket, or dilute/recycle. Jacket U for stirred liquids ≈ **300–1000 W/m^2K**; internal coils **500–1500**. Keep the **adiabatic-temperature-rise / runaway** margin honest: the batch cases `batch26_cooling_failure` and `batch27_semibatch_dosing` demonstrate thermal runaway when cooling can't keep up (`batch06_adaptive_runaway` no longer does: its esterification is nearly thermoneutral).

2.4 Sizing rules

- **Liquid CSTR:** τ from kinetics; vessel **H/D ≈ 1–3**, fill to **70–85%**, freeboard for disengagement. Confirm against `batch01_first_order`.
- **Catalytic fixed bed:** superficial gas velocity **0.3–1.5 m/s**; keep **ΔP $\leq$ 10–20%** of inlet P (Ergun); bed **L/D $\geq$ 5**, particle **3–6 mm** to balance ΔP vs internal diffusion.

- **Fluidised bed:** operate at **2–7× U__mf**, typically **0.3–0.7 m/s**; tall freeboard to cut entrainment.
- **Space velocity** sanity check: gas catalytic **GHSV 500–10,000 h^{-1}**; liquid **LHSV 0.5–5 h^{-1}**.

3 Distillation & separation-train design

Distillation is the workhorse: ~90% of liquid separations, but also the plant's largest single energy sink (reboiler duty). Every heuristic below is a **range to verify** — run the case, read the diameter and reflux the solver reports, and decide.

3.1 Internals selection (trays vs packing)

The *why*: trays handle fouling, solids and wide turndown; structured packing gives low pressure drop and high efficiency per metre but is intolerant of fouling and expensive.

Service / criterion	Pick	Typical numbers	Why
D > 0.8 m, clean, normal P	Sieve/valve trays	spacing 0.45–0.6 m, ΔP ≈ 0.7 kPa/tray	cheap, robust, wide turndown (valve 4–5:1)
D < 0.8 m	Random packing	HETP ≈ column D	trays uneconomic to install in small shells
Vacuum (P < 0.3 bar)	Structured packing	ΔP 0.1–0.4 kPa/m vs 0.7 kPa/tray	low ΔP keeps reboiler T (and thermal-degradation) down
High purity / many stages	Structured packing	HETP 0.3–0.5 m, a 250–350 m^2/m^3	most stages per metre of height
Fouling / solids / corrosive	Trays	—	cleanable, replaceable
Foaming, low ΔP	Random (Pall/IMTP)	HETP 0.5–0.9 m	open area tolerates surging

3.2 Diameter from vapour load (flooding)

Size on the controlling (usually top) vapour rate. Use the **C-factor / Souders-Brown** velocity: u_flood = C_sb · √((ρ_L−ρ_V)/ρ_V), C_sb ≈ 0.07–0.12 m/s at 0.45–0.6 m spacing. **Design at 75–85% of flood** (70% for foaming systems). Equivalently the **F-factor** F = u · √ρ_V should land at **1.2–2.5 (m/s)(kg/m^3)^0.5** for trays, up to ~2.5–3 for structured packing. Typical superficial vapour velocities: **0.5–1.5 m/s** at atmospheric, **2–4 m/s** under vacuum (low ρ_V). Then A = V/u, D = √(4A/π). Aspect ratio H/D ≈ 8–30; if H/D > 30 split into two shells. Weir loading and downcomer set the lower bound — turndown below ~50% of flood weeps. `column01_benzene_toluene` reports diameter and the flood fraction; halve the feed and watch it shrink.

Why the top and the bottom of a column want different diameters. The flooding diameter follows the **vapour VOLUMETRIC flow**, not the molar one, and the two sections differ in every term that turns one into the other: temperature, pressure, vapour molar mass and usually the vapour molar flow itself. Below the feed the liquid traffic is larger, which raises the flow parameter F_LV = (L/V)√(ρ_V/ρ_L) and therefore LOWERS Fair's capacity parameter C_sb — so a stripping tray floods at a lower velocity than a rectifying one. Whether that makes the bottom the wider section is not a rule: the vapour volumetric flow may fall faster than the capacity does, and then the top governs. It is decided by whichever tray loses the argument, tray by tray, which is exactly what `column09_tray_hydraulics` prints. It reports **1.21 m** for its rectifying section against **1.32 m** for its stripping section, on the same trays and the same spacing.

Swaging is an ECONOMIC decision, not a thermodynamic one. Having two diameters, you may build a **swaged** tower — a narrow section, a transition cone and a wide section — or one **straight** shell at the larger diameter. The straight shell buys plate you do not need; the swaged one buys a transition, its fabrication, its extra weld inspection and a second set of tray sizes to hold in stores. The usual rule of thumb: **swage only when the two sections differ by roughly 15–20% in diameter**, and build straight below that, because under about 15% the transition costs more than the plate it saves. Judge it on DIAMETER, not on area — 20% in diameter is already 44% in cross-section. The 8% gap in the case above is comfortably inside the band, so that tower is built straight at 1.32 m, and Choupo says so without deciding for you: it publishes both section diameters and their difference and prices no transition, because it has no correlation for one.

3.3 Reflux and FUG sizing

Shortcut design (Fenske-Underwood-Gilliland, `shortcut01_benzene_toluene`): Fenske gives N_min at total reflux, Underwood gives R_min, Gilliland interpolates. **Heuristic R = 1.1–1.5 · R_min** — the cost optimum, because below 1.1 N (and column height) blows up, above 1.5 reboiler/condenser duty and diameter grow with little stage saving. **N ≈ 2 · N_min** at that reflux. Feed near the stage matching its composition. Validate the shortcut against the rigorous MESH run (`column02_simultaneous`).

3.4 Sequencing multicomponent splits

Direct sequence (lights overhead first) dominates. Heuristics, in priority: (1) remove **corrosive/hazardous** components early; (2) do the **most-plentiful** product split early; (3) save the **hardest split** ($\alpha < 1.2$) and the **highest-purity** split for last, alone; (4) favour **near-50/50** splits; (5) "lightest-first" when relative volatilities are ordered. Each saves reboiler duty by not re-boiling already-separated material.

3.5 Azeotropes

When $\alpha \to 1$ or an azeotrope blocks the split, ordinary distillation cannot cross it. Wang-Henke (bubble-point) goes **unstable near azeotropes — switch to `method simultaneous`;** (rigorous MESH Newton, `column03_azeotrope_mesh`). Options: pressure-swing (if the azeotrope moves $\geq$5 mol% over the P range), entrainer/extractive, or hybrid with the membrane unit. Set ΔT_min $\geq$ 10 °C on the reboiler/condenser to keep utility cost sane.

4 Absorption, stripping & gas treating

Absorption transfers a gas-phase solute into a liquid solvent; stripping is the reverse. Reach for it instead of distillation when the species is DILUTE in a gas, or when no boil-up is available — you contact, you do not boil. **Hypothesis to run:** if the solute is < ~10 mol% of a gas and a solvent gives a low slope m = y/x, absorption is far cheaper than condensing and distilling the whole stream.

Situation	Why absorption/stripping wins
Dilute solute in a large gas stream	no need to condense/boil the bulk inert
Acid-gas removal (CO2, H2S)	selective chemical solvent (amine) near ambient T
Solvent / VOC recovery from a vent	recover below its dew point without refrigeration
Heat-sensitive solute	ambient contacting, no reboiler

The absorption factor and Kremser. The master group is $\mathbf{A = L/(m \cdot V)}$ (liquid rate / equilibrium-slope × gas rate); stripping uses $\mathbf{S = 1/A = mV/L}$. Design $\mathbf{A \approx 1.2–2.0}$: at $A < 1$ you can never reach a fixed recovery no matter how tall the tower; at $A > 2$ you waste solvent and the recovery curve flattens. The **Kremser equation** turns A and the target recovery into ideal stages — ~99% recovery at A = 1.4 needs roughly **6–10 ideal stages**.

Minimum solvent. The minimum L/V is where the operating line touches the equilibrium line; design at **L = 1.2–1.5 × L__min** — like reflux, less is infeasible and much more just inflates the downstream stripping duty. Always judge the absorber+stripper PAIR, never the absorber alone.

Internals	Pick when	Numbers
Random / structured packing	corrosive, foaming, low ΔP, small diameter (< ~0.6 m)	HETP 0.3–0.6 m (structured), 0.5–1 m (random)
Trays	large diameter, high liquid load, side-draws, fouling	0.3–0.6 m spacing, 60–80% stage efficiency

Quote packed height as **HTU × NTU** (HTU ~0.3–1 m); run the gas at **60–80% of flooding** (the same F-factor/superficial-velocity idea as distillation). **Gas treating:** for CO2/H2S use a chemical solvent (MEA/MDEA, 15–50 wt%) — the reaction boosts capacity at low partial pressure, but regeneration costs **~1–4 MJ per kg CO2** and dominates the operating cost, so solvent choice is really a regeneration-energy decision. Cold + high-P favours absorption; hot + low-P favours stripping — the same column run in reverse.

5 Liquid-liquid extraction

Extraction (LLE) separates by *chemical affinity* to a second liquid phase, not by volatility — so it wins exactly where distillation struggles. **Hypothesis to run:** take a feed and check whether the relative volatility α and the solvent distribution coefficient K point the same way; if $\alpha \to 1$ but K is large, LLE is the cheaper route.

When LLE beats distillation (the four classic triggers):

Trigger	Distillation symptom	Reach for LLE when
Close-boiling	$\alpha < 1.05$–$1.10 \to 80+$ trays, high reflux	a solvent gives $K \geq 5$–10 with selectivity $\beta > 20$
Azeotrope	pinch, can't cross	solute partitions selectively (aromatics/aliphatics, acetic acid/water)
Heat-sensitive	thermal degradation > 100–150 °C	ambient-to-40 °C extraction protects the product (antibiotics, vitamins)
Dilute aqueous	boil-up energy $\propto$ whole stream	solute < 5–10 wt% and $K \cdot (S/F) \gtrsim 5$ (metals, phenol, caprolactam)

The catch: LLE almost always needs a downstream distillation to *recover the solvent*, so judge the **combined** train, not the extractor alone.

Solvent selection — rank candidates on four numbers:

- **Distribution coefficient $K = y/x$** (solute in solvent / in raffinate). Want $K \gtrsim 5$–10; below ~1 the solvent flow balloons.
- **Selectivity β = K__solute/K__carrier**, the LLE analogue of relative volatility. $\beta > 1$ is mandatory; $\beta >$ 20–50 lets you reach high purity in few stages. $\beta \to 1$ is the LLE "azeotrope."
- **Recoverability**: a 10–30 °C boiling-point gap (or a density/solubility swing) between solvent and solute so the recovery still is cheap.
- **Density difference $\Delta\rho \geq$ 50–150 kg/m^3** and **moderate interfacial tension $\gamma \approx$ 5–30 mN/m** for clean settling — too low γ emulsifies, too high needs intense agitation.

Number of stages. Use the Kremser equation with the extraction factor $\mathbf{E = K \cdot (S/F)}$. Design for **E ≈ 1.3–2.0**: at $E < 1$ you can never reach high recovery regardless of stages; at $E > 3$ you are wasting solvent. For 99% solute recovery at $E = 1.5$, Kremser gives roughly **8–12 ideal stages**; raising E to 2 cuts it to ~5–7. Stage efficiencies are low (mixer-settler 75–95%, columns 20–40% per "stage"), so HETS dominates real height.

Solvent-to-feed ratio (S/F). Set by E and K: $\mathbf{S/F = E/K}$. With $K = 5$ and $E = 1.5$, $S/F \approx 0.3$; with a weak $K = 1$, the same E forces $S/F \approx 1.5$ — and a big solvent-recovery column. Stay in **S/F ≈ 0.2–2**; outside it the economics flip back toward distillation.

Equipment selection by throughput and settling tendency:

Contactor	Throughput	Stages (typ.)	Use when
Mixer-settler	low–very high	1 per unit, trains of 3–6	slow-settling, high efficiency, many stages, easy scale-up (footprint large)
Sieve-tray / packed column	medium–high	5–10	good $\Delta\rho$ & γ, few–moderate stages, low cost
RDC / pulsed / Karr (agitated)	medium	8–15	low γ, emulsifying systems, tall but compact footprint
Centrifugal (Podbielniak)	low–medium	3–7	tiny $\Delta\rho$, short residence (antibiotics), high cost

Why: agitation buys mass transfer but fights settling — the design tension is always *contact intensity vs. phase disengagement.* In Choupo, sweep S/F and stage count on an absorber/extractor case and watch the Kremser recovery curve flatten past $E \approx 2$ before committing solvent.

6 Membranes & aqueous/advanced separations

Membranes win where the separation is **isothermal and low-grade-energy-only**: aqueous streams, heat-sensitive or non-volatile solutes, and modest concentration ratios. The rule of thumb: **if the relative volatility is favourable and the solvent must be boiled anyway, distillation/evaporation wins; if you only need to *concentrate* an aqueous solution or strip ions/macromolecules at near-ambient T, a pressure-driven membrane is 5-20× cheaper in energy.** RO desalination runs ~3-4 kWh/m^3 vs ~10-15 kWh/m^3 (with vapour recompression) for thermal — because you pay only to overcome osmotic pressure (~0.78 bar per 1000 mg/L NaCl), not the latent heat of water.

6.1 Selecting the membrane class

Class	Pore / MWCO	ΔP (bar)	Flux (LMH)	Rejects	Use when
MF	0.1-10 µm	0.5-2	50-1000	suspended solids, bacteria	clarification, cell harvest, RO pretreatment
UF	1-500 kDa	1-5	30-150	macromolecules, colloids, virus	protein concentration/diafiltration, oily water
NF	200-1000 Da	5-25	15-60	divalents, sugars, dyes; passes monovalents	softening, sugar/whey, colour removal
RO	<200 Da	15-70 (BWRO 10-25, SWRO 55-70)	12-30	nearly all ions	desalination, final dewatering, ZLD

Driving pressure scales with what you reject: as MWCO falls, osmotic back-pressure and fouling resistance rise, so flux drops and ΔP climbs. NF is the "selective" sweet spot — it passes NaCl (low osmotic penalty) while holding SO_4^{2-}/Ca^{2+} (*membrane02_NF_sugar*).

6.2 Recovery vs scaling — the master trade-off

Higher single-pass recovery *r* shrinks capital and brine volume, but concentrates sparingly-soluble salts at the wall by the **concentration factor CF = 1/(1−r)**. At r = 0.75, CF = 4; at r = 0.90, CF = 10. Cap recovery so the **scaling saturation index stays below the threshold**: keep **LSI < 0 (or < +1.8 with antiscalant)** for $CaCO_3$ and **S&DSI / IAP-Ksp ratio < 0.8** for $CaSO_4$, $BaSO_4$, SiO_2 (<120 mg/L). This is exactly the recovery-vs-SI sweep in *optim04_membrane_recovery_scaling* and *membrane07_scaling_si / membrane08_softened_scaling* (softening lifts the ceiling). Typical r: **SWRO 35-50% per pass, BWRO 75-85%, NF 80-90%** with antiscalant.

6.3 Crossflow, flux and polarisation

Concentration polarisation multiplies wall concentration by **CP = exp(Jw/k)**; keep it **< 1.3-1.5**. Push the mass-transfer coefficient *k* up with crossflow: spiral-wound feed-channel **superficial velocity 0.1-0.4 m/s** (Re 100-1000, spacer-filled), tubular/UF **2-6 m/s**. Design flux conservatively below the **critical/limiting flux**: **SWRO 12-17 LMH, BWRO 18-27, NF 15-40, UF 30-100**. Over-fluxing collapses *k* and locks in irreversible fouling. *membrane03_concentration_polarization* and *membrane04_spacer_schock_miquel* let you watch CP rise as velocity drops.

6.4 Staging, trains, pretreatment

Spiral-wound elements are 8”×40”, ~37 m^2 each, 6-8 per pressure vessel. To hold velocity as permeate is removed, taper the **array (e.g. 2:1 → 6 then 3 vessels)**; add a **second pass** on permeate when you need <500 mg/L product. Run a tapered train in *membrane05_train*. Budget **per-vessel ΔP 1-3 bar** (replace at 15% rise). Always pretreat: MF/UF or coagulation to **SDI_{15} < 3**, antiscalant, and cartridge filtration — fouling, not membrane area, sets RO lifetime.

7 Solids processing & particle technology

Solids units are governed by the **particle**, not the molecule: crystal habit, size distribution and mechanical strength dictate every downstream choice. Always carry a target median size (d50) and span as design specs.

7.1 Crystallisation route — pick by the solubility curve

The route follows how solubility varies with the achievable handle. Generate supersaturation (S = c/c*) gently: hold the **metastable zone width**, typically $\Delta c/c \approx$ *0.01–0.05** (S ≈ 1.01–1.05), so growth dominates nucleation and you get coarse, filterable crystals. Push S past ~1.1 and you nucleate a fine, hard-to-filter slurry.

Route	Use when	Driving handle	Typical yield knob	Watch
Cooling	dc*/dT steep (>1 wt%/10 °C, e.g. KNO_3, KCl)	ΔT, 0.1–1 °C/min ramp	ΔT over solubility curve	wall encrustation; keep ΔT_wall<5–10 °C
Evaporative	flat solubility (NaCl, ~0.07 wt%/°C)	solvent boil-off	evaporation rate	scaling, foaming; vacuum to cut T
Antisolvent	high solubility, T-insensitive; heat-sensitive API	miscible non-solvent addition	solvent:antisolvent ratio	local high-S → fines; add slowly, well-mixed
Reactive/precip.	sparingly soluble product	reagent feed rate	stoichiometric excess	very fast nucleation → nanoparticles

MSMPR sizing: mean residence τ = **1–4 h**; the population balance gives d43 $\propto G\tau$, so longer $\tau \Rightarrow$ bigger crystals at fixed growth rate G (~10^{-8}–10^{-7} m/s). See `crystalliser02_msmpr` (population balance), `crystalliser06_nacl_antisolvent` and `crystalliser07_nacl_antisolvent_cooled` for combined routes.

7.2 Solid–liquid separation — filtration vs centrifugation

Pick by particle size and solids fraction. Vacuum/pressure filtration suits **d50 > 10–20 µm**; below ~5 µm cake resistance explodes (specific cake resistance α rises) and you switch to centrifugation or add filter aid. Sedimenting centrifuges handle $\Delta\rho$ > **50 kg/m³** and feeds too fine or too compressible to filter; decanters take 5–40 wt% solids, disc-stacks <5 wt%. Rule: filter when cake washing/dryness matters and particles are coarse; centrifuge when fines, fragile crystals, or fast throughput dominate.

7.3 Dryer selection — by feed state and particle size

Dryer	Feed	Product d50	Residence	Notes
Spray	pumpable slurry/solution (<40% solids)	10–200 µm	5–30 s	hollow spheres; high T_gas but short → heat-sensitive OK. `sprayDryer01_sugar`
Fluid-bed	free-flowing granules	50 µm–5 mm	10–60 min	narrow size range; needs U above U_mf
Drum	viscous paste/slurry	flakes	5–30 s	conductive, no carrier gas
Tray/batch	small lots, fragile	any	2–24 h	cheap, labour-heavy

`solidDryer01_sugar` shows a through-circulation solid dryer; `sprayDryer02_residence_sweep` sweeps τ vs moisture.

7.4 Gas–solid separation & fluidisation velocities

Cyclones are the workhorse for **d > 5–10 µm**; below that, efficiency falls and you add a bag filter (`bagFilter01_dust`). Design the **inlet velocity to 15–25 m/s** (>30 m/s re-entrains/erodes; <10 m/s drops cut size); body-to-inlet ΔP ≈ 0.5–2 kPa, 4–8 inlet velocity heads. See `cyclone01_dust_removal`, `cyclone02_leith_licht`.

Superficial-velocity targets: **fluid-bed operate at 2–4×U_mf** (above minimum fluidisation, below terminal U_t to avoid carryover); **pneumatic conveying dilute-phase 15–25 m/s**; settling-chamber gas <3 m/s. Always check the actual U_mf and U_t from particle d and $\Delta\rho$ rather than trusting the ratio blindly — that is the glass-box hypothesis to run.

8 Evaporation & multiple-effect

Evaporation concentrates a solution by boiling off solvent (usually water). The whole game is **steam economy** — kg water evaporated per kg live steam — and the way to raise it is to reuse each effect's vapour as the next effect's heating steam. **Hypothesis to run:** sweep the number of effects on Choupo's triple-effect case and watch economy climb while area and capital grow.

Effects	Steam economy	Pick when
1	0.8–0.95	small duty, cheap steam, scaling/fouling service
2–3	1.7–2.6	medium duty — the common industrial choice
4–6	3.5–5.0	large water removal, expensive steam (sugar, paper, desal)

Each added effect adds roughly **+0.8 kg evaporated per kg steam**. WHY it saturates: the available ΔT (steam T minus the vacuum/condenser T) is **split** across the effects, so each has a smaller driving force and needs more area. Capital $\propto$ effects while the economy gain shrinks — the optimum is usually **2–5 effects**.

Feed arrangement	Direction	Use when
Forward	feed with steam	feed near boiling; simplest, no inter-effect pumps
Backward	feed counter to steam	cold or viscous feed (the hottest effect handles the thickest, lowest-viscosity liquor)
Mixed	hybrid	balance viscosity and economy

Boiling-point elevation (BPE): dissolved solids raise the boiling point above pure water — a few °C when dilute, **15–25 °C** for strong NaOH or brine — eating into the ΔT and capping the last effect (Choupo's NaOH and sugar `evaporator` cases account for it). **Type selection:** falling-film for low-ΔT, heat-sensitive, low-viscosity service (high U); **forced-circulation** (pumped 2–3 m/s) for viscous, scaling, or crystallising liquors; agitated/wiped-film for very viscous. **Vapour recompression** (electric MVR, or TVR with a steam ejector) recycles the vapour's latent heat into the same effect — one MVR effect can match a 4–6-effect train when electricity is cheap relative to steam.

9 Gas-liquid separators & process drums

Every flash, every reflux accumulator, every compressor suction line needs a vessel that lets vapour and liquid part ways. The physics is a settling race: a liquid droplet falls under gravity while the rising vapour drags it up. Size the vessel so the vapour goes slow enough that droplets above a cut size (~150 µm bare, ~10 µm with a mesh pad) rain out instead of carrying over.

The Souders-Brown anchor. The maximum allowable vapour velocity is `v_max = K·sqrt((ρ_L - ρ_V)/ρ_V)`. The square-root group is just the terminal-settling balance (gravity vs. drag); `K` (m/s) bundles droplet size, drag coefficient and a safety margin. The cross-section follows from `A = Q_V/v_max`, so K sets the diameter directly — get it wrong and you either spray liquid downstream or buy steel you didn't need.

K-factor (m/s)	Service / condition	Why
0.03–0.06	No demister, bare gravity drum	Only ~150 µm+ drops settle; derate hard
0.06–0.07	Vacuum / low P (< 1 bar)	Low ρ_V makes the group large; cap velocity
0.08–0.11	Mesh demister pad, clean service	Pad coalesces 5–10 µm drops; run faster
×0.7–0.8	Foaming / fouling / glycol-amine	Foam re-entrains; cut K by 20–30 %
×0.9 per	0.5–1.0 m of disengagement below pad	Short disengaging height steals settling time

Use the **vapour density at operating T,P**, not standard — a compressor suction drum at 5 bar has ~5× the ρ_V of atmospheric, shrinking `v_max` by ~2.2× and demanding more area than newcomers expect.

Vertical vs. horizontal. Choose by vapour-to-liquid loading and footprint.

Pick	When	Why
Vertical	High vapour / low liquid; compressor **suction & K.O. drums**; tight plot	Settling distance = height is cheap to add; small liquid hold
Horizontal	High liquid load, large surge/holdup, **reflux accumulators**, slug catchers, 2 liquid phases	Big interfacial area per volume; long L for L-L decant; lower vessel

For horizontal drums the disengagement area is `L·(D minus liquid level)`, so set the vapour space to ~40–50 % of the cross-section and check `v_max` against that reduced area.

Geometry and holdup. Aim for **L/D ≈ 3–5** (below 2 the heads dominate cost; above 6 it gets whippy and hard to support). Liquid **residence time 5–15 min** of working volume sets the length/level: ~5 min for a clean reflux drum feeding a good level controller, 10–15 min where downstream is a fired heater or compressor you must never starve or slug. Allow **vapour disengagement height ≈ 0.3 · D or 0.45 m (whichever larger)** between the high liquid level and the inlet/demister, plus ~0.15 m mesh pad. Inlet nozzle momentum $\rho \cdot v^2$ should stay **< 2250 kg/m · s^2** (< ~1000 if no inlet device) or the jet shatters liquid faster than the pad can catch it.

Glass-box hypotheses to run. In Choupo a flash drum is just an `IsothermalFlash` at (T,P); the vapour and liquid rates it returns are your `Q_V` and holdup feed. Try `tutorials/steady/flash/flash01_benzene_toluene`: take the V and L rates, pick K = 0.08 (pad) vs. 0.04 (bare), and watch the required diameter roughly double when you drop the demister and halve K. Then re-flash at 5 bar and confirm the higher ρ_V forces a smaller `v_max` and a fatter drum — the Souders-Brown sqrt made visible.

10 Heat exchangers — type selection & sizing

Pick the *type* first — it sets cost, U, and ΔP before you ever size area. The driving question is the **overall coefficient U**: a gas film (h ≈ 30–300 W/m^{2}K) starves U, so the cheap-per-m^2 geometry wins; a clean liquid–liquid duty (h ≈ 1000–5000) lets you chase compactness and low ΔT_min.

10.1 Type-selection table

Type	Best service	U (W/m^{2}K)	ΔP budget	T/P limit	Why
Shell-and-tube (TEMA)	Any phase, high P/T, fouling	300–1200 (liq-liq), 30–300 (gas)	0.3–0.7 bar	<40 bar, <600 °C	Rugged, mechanically cleanable, ASME default
Plate-and-frame (gasketed)	Clean liquid–liquid, close approach	2000–7000	0.3–1 bar	<25 bar, <180 °C	Huge area/volume, ΔT_min ≈ 1–5 °C; gaskets limit T
Air-cooled (fin-fan)	Hot streams, no cooling water	15–60 (bare-tube basis)	gas side ~0.005 bar	tube-side any	Saves water; approach to ambient ≥ 10–15 °C
Double-pipe	Small duty (<20 m^2), high P, dirty	300–1000	flexible	very high P	Cheap, true counter-current (F=1)
Spiral	Slurries, fouling, viscous	700–2500	self-scouring	<20 bar	Single channel resists plugging
Printed-circuit (PCHE)	Very high P, compact (LNG, H_2)	1000–5000	tight	<600 bar	Diffusion-bonded; near-zero fouling tolerance

10.2 Gas-phase service

Shell-and-tube *still* makes sense for gas — but put the **gas on the shell side** with the gas film controlling, accept U ≈ 30–300, and expect large area. For sensible gas cooling against ambient, **air-cooled** usually beats shell-and-tube + cooling-water tower on TAC once duty > ~0.5 MW. For gas–gas (e.g. feed/effluent), a plate-fin or PCHE recovers heat that a shell-and-tube cannot economically reach.

10.3 Allocation & velocities

Put on the **tube side**: the dirtier, higher-pressure, more-corrosive, or lower-flow stream (easier to clean, cheaper to alloy small tubes).

- Tube-side liquid: **1–2.5 m/s** (>1 m/s suppresses fouling; <3 to limit erosion; <1.5 for SS to avoid erosion-corrosion).
- Shell-side liquid (cross-flow): **0.3–1 m/s**.
- Gas, tube or shell: **10–30 m/s** (process gas), up to 30 m/s for low-density vapour; chase the gas film, watch ΔP.
- Condensing vapour: keep below ~50% of erosion velocity; flooding governs.

10.4 Phase-change service

- **Condensers**: prefer shell-side condensation (vapour on shell, coolant in tubes); U ≈ 800–1500 for organics, 1500–4000 with steam. Total condenser → ΔT driven; allow vapour velocity de-rating.
- **Reboilers**: kettle (U ≈ 800–1200) for wide turndown and dirty bottoms; thermosiphon (U ≈ 1000–1500) cheaper but needs ΔT > a few °C and stable circulation. Cap heat flux below the **critical ~30–40 kW/m^2** to dodge film boiling. See Choupo `tutorials/steady/distillation` (condenser/reboiler duties) and `tutorials/steady/heat`.

10.5 ΔT_min, LMTD correction, fouling

- **ΔT_min**: 10–20 °C for hot utility/gas, 5–10 °C liquid–liquid, 3–5 °C refrigeration (energy vs area trade-off — verify via Choupo pinch composite curves).
- **LMTD correction F**: keep $\mathbf{F > 0.8}$; $F < 0.75$ means add shell passes (2-4, 4-8) or go pure counter-current.
- **Fouling**: add R_f ≈ 0.0002 (clean liquid) to 0.0005–0.001 m^2K/W (cooling water, fouling organics); this can dominate 1/U and is why a glass-box run should report the clean-vs-fouled area split.

11 Heat integration & pinch analysis

Heat-exchanger SELECTION sizes one match; pinch analysis decides which streams should match AT ALL and sets the energy target before any network is drawn. It is the highest-leverage utility-saving tool — typically **20–30% off the energy bill** versus a naive design. **Hypothesis to run:** build the composite curves in Choupo's Pinch view and read the targets before placing a single exchanger.

Composite curves and the pinch. Combine all hot streams into one composite (T vs $\Sigma Cp \cdot \Delta T$) and all cold streams into another; slide them together until the closest vertical approach equals **ΔT_min**. That point is the **PINCH**; the overshoots are the minimum hot and cold utilities (the **MER targets**), fixed before design.

Service	ΔT_min	Why
Liquid–liquid	10 °C	default; balanced area vs energy
Process with a gas stream	20–40 °C	low U → area explodes at small ΔT
Refrigerated / cryogenic	3–5 °C	cold is expensive; pay area to save it
Near a furnace / high-grade heat	20–60 °C	high-grade heat is cheap relative to area

Supertargeting plots total cost vs ΔT_min and picks the flat minimum (often **8–15 °C** for fluids).

The three golden rules — break any and you spend MORE than the target:

1. **No heat across the pinch** — a cross-pinch exchanger adds the same amount to BOTH utility bills.
2. **No hot utility below the pinch** — below it there is a heat surplus; adding heat just needs more cooling.
3. **No cold utility above the pinch** — above it there is a deficit; cooling there just needs more heating.

The grand composite curve (GCC) plots surplus/deficit vs shifted T: read it to place utilities at the RIGHT level — pick the cheapest steam/refrigerant the pockets allow, spot an intermediate utility (cheaper MP steam, or raising steam from recovered heat), and see where a fired heater or molten salt is GENUINELY needed versus where recovered heat suffices.

12 Fired heaters & furnaces

Reach for a fired heater when the duty is large (>3 MW) or the process temperature is above what your hottest utility can deliver. HP steam tops out near 250 C and 40 bar; HiTec molten salt reaches ~535 C; Dowtherm A ~400 C. Above that — crude-unit charge heaters (350-400 C), reformer feed, vacuum-tower heaters, reboilers on high-boiling bottoms — you must burn fuel directly, because the WHY is simple: a flame radiates at 1500-1900 C, so it can drive heat across a tube wall that no recirculated fluid can. Below ~535 C and a few MW, molten salt or fired-salt loops usually win on safety and turndown; don't fire directly if a utility reaches the target.

Two heat-transfer zones, by mechanism. The RADIANT section (firebox) absorbs 60-75% of the duty by flame radiation; tube-wall view factor and flame emissivity dominate, so the design knob is *flux*, not area. The CONVECTION section above it scavenges 25-40% of the duty from 700-900 C flue gas by convection over finned/studded tubes, dropping stack gas to 150-250 C. The split exists because radiant flux is capped: push too hard and you coke the film.

Heater type	Duty range (MW)	Typical use	Radiant flux (kW/m2)	Why pick it
Vertical-cylindrical	0.5-20	Small/medium, plot-limited	25-35	Cheapest, smallest footprint, uniform firing
Box (cabin), horizontal tubes	10-100	Crude/vacuum charge, reboil	30-40	Big duty, even flux, many burners
Box, two-fired cell	50-300+	Reformers, ethylene cracking	35-50 (cracking, short residence)	Highest duty, double-fired uniformity
Cylindrical helical-coil	0.5-10	Hot-oil / salt heaters	20-30	Compact, low-fouling clean service

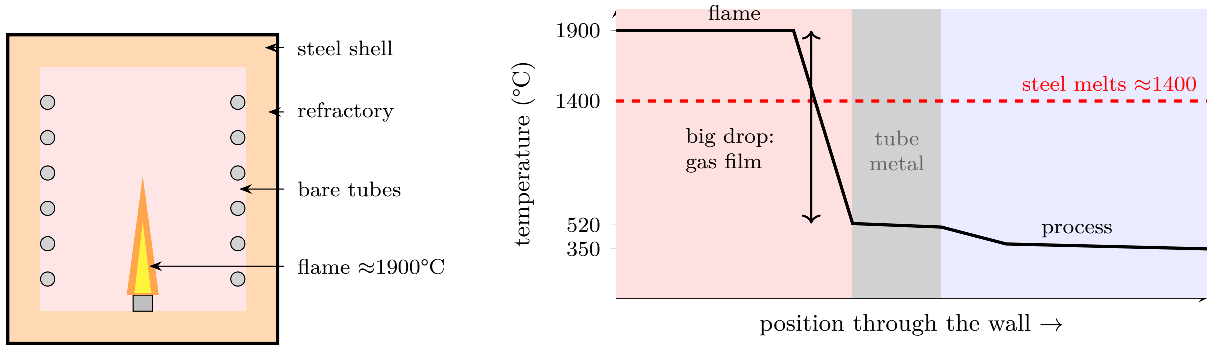
Radiant flux — the master rule. Design average radiant flux 30-40 kW/m2 (some references 25-45). Go higher and the *peak* local flux — typically 1.5-2x the average — drives the tube-wall film over its coking threshold (~430 C for crude oil, lower for residue), laying down coke that insulates, raises tube metal temperature, and shortens runlength. Average flux is therefore a runlength decision, not just an area decision: halving flux can double time-between-decokes.

Efficiency, excess air, draft. Thermal efficiency 75-82% bare; 85-92% with an air preheater recovering stack heat. Each 20 C of stack-gas drop is worth ~1% efficiency, and 1% excess-air reduction another ~0.5%. Run 10-15% excess air on gas, 15-25% on oil — enough to complete combustion, not so much you heat nitrogen up the stack. Draft: natural (buoyant stack, simplest, cheap) for small heaters; forced or balanced draft once you add an air preheater (the preheater's pressure drop demands a fan). Keep a slight negative draft (-2 to -5 mm H2O) at the arch so flue gas leaks in, not flame out.

Tube metallurgy sets the limit. Specify tubes by metal temperature = process T + flux-driven ΔT (50-100 C) + fouling margin: carbon steel to ~400 C, 1.25Cr-0.5Mo to ~550 C, 9Cr-1Mo to ~650 C, HK/HP cast austenitics to 1000 C+ for reformers. NOx scales with flame temperature — low-NOx burners (staged air/fuel, flue-gas recirculation) cut it to 30-50 ppm but lengthen the flame, so check flame-to-tube clearance. Hypothesis to run in Choupo: model the heater as a W-equivalent duty against a fired-fuel stream, then sweep excess air and stack temperature to watch efficiency and fuel firing rate move — confirm each 20 C stack drop buys ~1%.

Why the tubes do not melt at a 1900 °C flame. Students reasonably ask how steel survives a flame far above its melting point. The answer: **the tube never sees the flame temperature.** The process fluid races through the inside and carries the heat away continuously, so the tube wall sits at roughly **process-T plus a small film rise** — typically **400–600 °C metal**, not 1900 °C. The radiant flux (30–40 kW/m^2) is balanced by the inside film: the film rise = flux / h_inside is only **30–120 °C** for a fast liquid (h $\approx$ 1000–3000 W/m^2K). The flame's heat is real, but it is removed as fast as it arrives.

Two lessons: (1) the metallurgy is specified for the **tube-metal temperature**, not the flame (carbon steel ~400 °C, 9Cr-1Mo ~650 °C, cast HK/HP austenitics to ~1000 °C); the firebox walls are protected by **refractory** lining, not bare metal. (2) If the process flow STOPS or drops (pump trip, coking), the heat is no longer removed and the tube temperature runs away in **seconds** — the tube ruptures. That is why a fired heater carries a **low-flow trip** and the flux is capped: a hot tube with no flow is the classic furnace failure.


steel shell
refractory
bare tubes
flame ≈1900°C
flame
1900
1400
520
350
temperature (°C)
steel melts ≈1400
big drop:
gas film
tube
metal
process
position through the wall →

(a) firebox cross-section

(b) temperature profile

Figure 1: Why furnace tubes survive a 1900 °C flame. (a) The radiant tubes hang BARE along the firebox walls, exposed to the flame; the refractory lines the WALLS (protecting the steel shell and re-radiating heat), not the tubes. (b) Almost the whole temperature drop falls across the outer gas film, so the tube metal sits near 520 °C – far below its melting point – because the process fluid inside removes the heat as fast as it arrives. Lose the flow and the metal races toward the flame temperature.

13 Compressors, expanders & pressure change

Gas pressure change is governed by *volumetric flow at suction* and *required pressure ratio* `r = P_out/P_in` — not mass flow. Because gas density falls as you compress, machine choice tracks **actual inlet $\mathbf{m^3}$/h** and `r`, then degrades to the next type as flow rises or `r` falls. Treat every heuristic below as a hypothesis: run the case in Choupo, read `W_shaft` and discharge T, and check you stayed inside the box.

13.1 TYPE SELECTION

Type	Inlet flow (actual m^3/h)	r per body	Typical η_poly	Why / physics
Fan / blower	10^3–10^6	<1.1 (fan), <2 (blower)	0.55–0.75	Low Δp; cheap; r limited before it becomes a "compressor"
Centrifugal	1,700–340,000	1.5–3 per wheel, ~10 multi-stage	0.75–0.85	Dynamic; smooth, oil-free, high flow; surge limits turndown
Axial	>65,000	1.2–1.5 per stage	0.80–0.88	Highest flow, best η; narrow operating window
Reciprocating	<16,000	up to 5–10 per stage	0.75–0.90	Positive-displacement; best for high r, low flow, high P (>200 bar)
Rotary screw	<5,000	3–4 (up to ~8 oil-flooded)	0.70–0.78	Robust, dirty/wet gas, modest r; oil carryover

13.2 Stages, ratio per stage & intercooling

Split `r` across N stages with **equal ratio per stage** `r_stage = r^(1/N)`, kept to ~**2–4** (cap ~4 for recips, ~3 for centrifugals). Equal ratios minimise total shaft work and equalise per-stage discharge T. The driver of staging is the **discharge-temperature limit**: keep `T_out` $\lesssim$ **`150-175 °C`** (lube-oil coking, gasket and rod-packing life; ~200 °C absolute ceiling). Adiabatic `T_out = T_in·r_stage^((γ-1)/(γη))`, so a single jump from 1→16 bar on a $\gamma\approx1.3$ gas blows past 250 °C — unacceptable. **Intercool back to ~40 °C** between stages: this cuts the next stage's volumetric work (you compress colder, denser gas) and typically saves 10–20 % total power for `r > ~5`. Pair an intercooler with a knock-out drum — cooling condenses water/heavy ends that would erode the next stage. In Choupo, model each stage as a `compressor` (W_shaft + η) feeding a cooler + flash; cooling-water or chilled-water utilities by T-level supply the intercooler duty.

13.3 Efficiency, surge & turndown

Quote **polytropic** efficiency (0.75–0.85 centrifugal, up to 0.88 axial) for design — it is path-independent and additive across stages, whereas isentropic η drifts with `r`. Multiply by ~0.95 mechanical to get brake power. Dynamic machines **surge** at ~60–70 % of design flow (flow reversal, violent oscillation), so usable **turndown is only to ~70 %** without recycle/anti-surge; recips turn down to ~25 % via clearance pockets or speed. Size with ~10 % flow margin above the surge line.

13.4 When an expander pays back

If a hot, high-P stream must be let down (≥3–4 bar drop, ≥a few hundred kW recoverable), replace the throttle valve with a **turbo-expander** (model as `turbine`, W_shaft + $\eta\approx$0.80–0.85). Recovered shaft work directly drives a generator or a companion compressor (common in refrigeration, FCC flue-gas, cryogenic N_2). The trade-off: expansion chills the stream (Joule–Thomson + work extraction), so verify you don't cross a dew/freeze point — a glass-box run shows the outlet T before you commit hardware.

14 Drivers: electric motors, steam & gas turbines

A pump or compressor is only as available as the machine that turns it. The *driver* is a separate selection from the hydraulic/thermodynamic duty: pick it by shaft power, by what HP steam or fuel gas the plant already has, and by what must keep running when the grid trips. In Choupo the rotating unit carries `W_shaft + eta`; the driver question is *where that shaft power comes from* and at what reliability.

Rule — electric motor is the default below ~0.5–1 MW. Standard TEFC induction motors are cheap, compact, ~93–97% efficient, and need no auxiliaries. WHY the ceiling: above ~1 MW you move to medium-voltage (4.16 kV+) switchgear and the incremental $/kWofmotor+variable-frequencydrive(VFD)climbs, whileasteamturbinethat* also * letsHPsteamdowntoausefullevelstartstowinoncombinedsteam - balanceeconomics.Hypothesistorun : sizethemotorat$ 110 − −1250.06–0.10/kWh.

Rule — steam turbine when HP steam exists, for very large duty, or as a trip-proof spare. A back-pressure turbine taking HP→MP/LP steam "pays" for shaft work by letting steam down a pressure level you needed anyway — marginal cost is only the lost power-generation credit, so isentropic efficiency of 65–80% (small) to ~85% (large) is acceptable. A condensing turbine (exhaust to ~0.1 bar) maximises work but dumps ~2 MJ/kWh of latent heat to cooling water. Critically, a steam turbine keeps spinning on a grid blackout, so the boiler-feedwater pump and reactor-quench pump are classically *steam-turbine-driven spares* on a motor-driven main (or vice-versa).

Rule — gas turbine only for very large power + heat (combined heat-and-power). Below ~5 MW the heat-rate and capital rarely justify it; above ~10–15 MW a gas turbine with a heat-recovery steam generator delivers shaft power *and* the HP steam header — the cogeneration case Choupo models with steam utilities by T-level.

Driver	Sweet-spot power	Efficiency	Pick it WHEN	Watch-out
Electric motor (fixed-speed)	< 0.5 MW	93–97%	small/medium duty, throttled or constant flow	dumps control margin across a valve
Electric motor + VFD	0.1–5 MW	90–95% (incl. drive)	flow varies > ±20%	drive cost, harmonics, ~3–5% drive loss
Back-pressure steam turbine	0.5–15 MW	65–85% isentropic	HP steam available, want MP/LP let-down	power tied to steam demand
Condensing steam turbine	1–30 MW	70–85% isentropic	large duty, run-on-trip spare	condenser + cooling-water latent load
Gas turbine (CHP)	> 10 MW	30–40% (≈80% w/ HRSG)	huge power + process steam	fuel-gas spec, NOx, hot-day derate

Rule — variable speed beats throttling when flow turndown exceeds ~±20%. Power scales ~speed^3 on a centrifugal machine, so trimming flow by slowing the driver saves far more than burning the surplus head across a control valve. WHY not always: a VFD adds ~3–5% standing loss and capital, so for near-constant flow a fixed-speed motor + control valve is cheaper and simpler.

Rule — spare to n+1 on critical service. Install one extra parallel machine (e.g. 3×50% pumps, run 2) so a single failure or maintenance outage does not stop the plant; mix drivers (one motor, one turbine) so a *common-mode* loss — grid or steam — still leaves one runner. Verify by toggling a unit offline and re-running the flowsheet: does the remaining capacity still meet duty?

15 Fluid transport, line sizing & hydraulics

Moving fluid costs capital (pipe + driver) and power (friction). Every rule below trades a *bigger pipe / lower velocity* (more steel, less ΔP, less erosion) against a *smaller pipe / higher velocity* (cheap, but pumping power and noise climb as ~$v^2 \cdot {}^8$). Pick the driver by **pressure-ratio first, then flow**.

15.1 Driver selection by pressure ratio

Service	Δp / ratio	Pick	Why (glass-box)
Incompressible liquid	any head	**Pump** (centrifugal <~3500 m, recip/PD for low-flow high-head, viscous, or metering)	$\rho \approx$const → head, not ratio, is the variable
Gas, ΔP < 0.03 bar	ratio ≈1.0–1.03	**Fan**	ρ change negligible, treat as incompressible
Gas, 0.03–3 bar	ratio ~1.03–1.3	**Blower** (one stage)	mild heating, no intercool
Gas, ratio > ~1.3	up to ~3–4/stage	**Compressor** (centrifugal; recip for very high ratio/low flow)	T_out rises with ratio^((γ−1)/γ) → **stage + intercool if total ratio > ~4**

Multi-stage rule: keep per-stage ratio ≈ (total ratio)^(1/N) so each stage does equal work and discharge T stays below ~150–200 °C (lubricant/material limit). See `tutorials/steady/rotating/compressor01_air` and the multi-stage logic behind `compressorTurbine01_turbo_booster`; pumps in `pump01_water` / `pump02_pressure_spec`.

15.2 Recommended superficial velocities

Line / service	v (m/s)	Note
Pump **suction** (liquid)	0.5–1.5	low to protect NPSH; ΔP_suction is precious
Pump **discharge** / process liquid	1.5–3 (up to ~4.5)	economic optimum; >4 erosion-corrosion risk
Liquid, viscous/slurry	0.5–1.5	keep below settling; above abrasion limit
Gas / vapour, atmospheric–low P	15–30	density-limited
Gas, high pressure	10–20	ρ high → same ρv^2 at lower v
Steam (sat./superheat)	20–40 (to 60)	erosion + wet-steam wire-drawing
Two-phase / flashing	10–25	avoid slug/annular; check $\rho v^2 <$ ~6000 $\mathrm{kg/m \cdot s^2}$

The economic optimum tracks **constant ρv^2** (kinetic head), which is why dense fluids run slower. A quick screen: **v_econ ≈ 2 m/s liquid, 20 m/s gas**, then verify ΔP in `tutorials/steady/hydraulics/pipe01_water_line`.

15.3 Economic pipe diameter & ΔP budget

- Rough optimum: **D_econ ≈ 0.26 · ṁ^0.5 · ρ^—0.37** (m, SI) for turbulent steel — the balance of pipe capital vs pumping power.
- **ΔP budget:** liquid lines **0.2–0.5 bar per 100 m** (≈0.5 kPa/m); gas lines **0.1–0.2 kPa/m**; long pipelines push lower (0.05). Total pump head = static lift + friction + equipment + **control-valve allowance**.
- **Control-valve ΔP:** allocate the *larger* of **20–33 % of dynamic (friction) ΔP** or ~**0.7 bar (10 psi)**, so the valve keeps authority at high flow. Too little → loop goes open-loop; too much → wasted head.

15.4 NPSH (the pump killer)

Require **NPSH_available − NPSH_required ≥ 0.5–1 m** margin. NPSH_a = (P_suction − P_vap)/ρg + static head − suction friction. Levers: raise the suction vessel, fatten/shorten suction line (low v, see table), subcool, or cut speed. Hot/near-boiling liquids (reboiler bottoms, deaerator) are the classic cavitation traps — design suction generously.

Always run it: these are *hypotheses* — size the line, then let Choupo print the actual ΔP, T_out and head so the chosen velocity and driver prove themselves.

16 Control valves, pressure letdown & choking

A control valve is a deliberate, adjustable restriction — you spend pressure to gain control. Two questions fix the hardware: which valve body and trim, and whether the drop is so large it CHOKES.

Valve	Rangeability	Use when
Globe (single / multi-stage trim)	20–50:1	throttling — the default; high ΔP; anti-cavitation / low-noise trim available
Ball / segmented	100–300:1	high rangeability, slurries, on-off-to-throttle
Butterfly	20–50:1	large lines, low ΔP, cheap

Pick the **inherent characteristic** to linearise the loop: **equal-percentage** when most of the system ΔP is elsewhere (the common default — valve gain rises with flow to offset its falling ΔP); **linear** when valve ΔP dominates; **quick-opening** for on-off. Liquid sizing (non-flashing): Q = Cv · sqrt(ΔP/SG); size for ~80% open at max flow and ~20% at min, so the plug never sits on the seat (no control) or wide open (no margin).

Choking — the answer to "is a big letdown done in one stage?"

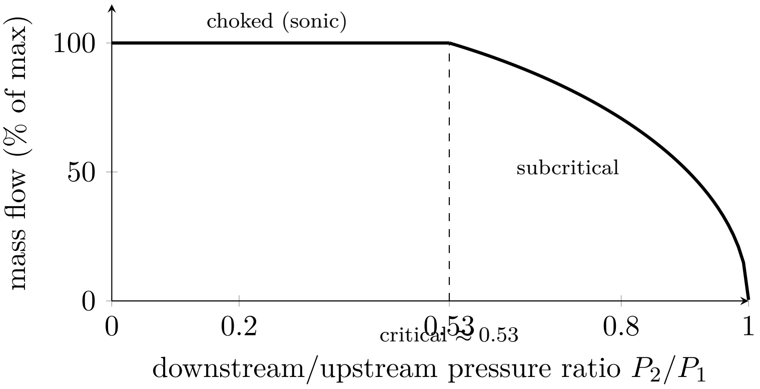


Figure 2: Gas flow through a valve vs the pressure ratio. Below the critical ratio (≈ 0.53 for air) the flow chokes – sonic at the vena contracta – and mass flow stops rising as P_2 falls; the surplus pressure drop becomes noise and erosion.

For a GAS, flow does NOT keep rising as you drop the downstream pressure. Once the pressure ratio falls below the **critical ratio** — r_c = [2/(γ+1)] raised to [γ/(γ−1)], ≈ **0.5 for most gases** (0.53 for air) — the velocity at the vena contracta reaches the speed of sound and the valve **CHOKES**: mass flow saturates, and the surplus ΔP buys no flow — it is dissipated downstream of the vena contracta in shocks and turbulence, as **noise (often > 110 dB), vibration and erosion**. Discharging 50 bar → 1 atm is a ratio of 0.02, i.e. deeply choked.

So a single standard valve CAN pass the flow, but across a deeply choked drop it screams, erodes its trim and shakes the pipe — which is why a large letdown is usually **staged**:

Letdown situation	Do this
Small gas ΔP, sub-critical	one globe valve
Large gas ΔP, choked	multi-stage / labyrinth trim, or valves in series, + a downstream silencer (each stage takes a sub-critical drop)
Liquid, P_2 and the vena contracta both > Pvap	standard trim
Liquid, P_2 > Pvap but the vena contracta < Pvap (cavitation)	imploding bubbles pit the trim — anti-cavitation (multi-step) trim, or an orifice downstream to raise the valve's backpressure; harden it

Letdown situation	Do this
Liquid, $P_2 < \text{Pvap}$ (flashing)	the outlet state requires the vapour, so no trim, staging or orifice removes it (an orifice only moves the two-phase crossing to the last device) — harden the trim, use an angle body, put the valve at the receiving vessel or flash on purpose into a drum

Rule of thumb: stage it when the predicted noise exceeds ~85–90 dBA or the trim-exit velocity is high. The glass-box check: drop the pressure across a `valve` in Choupo and read the outlet T and phase — a flashing liquid shows up as a vapour fraction at the outlet before you ever buy the trim.

17 Refrigeration & vacuum systems

Cold and vacuum are *bought power*. Both reject heat or pull gas against the ambient, so the design instinct is the same: **climb only as far down the ladder as the duty forces you, because every rung multiplies the shaft work.**

17.1 Refrigeration — pick the warmest level that works

Cooling water (CW) leaves a tower at ~30 C and can chill a process stream to ~35-40 C (10 C approach). The moment you need a process temperature **below ~40 C you have left the free zone** and pay for a compressor. The reason is the reverse-Carnot ceiling: ideal COP = T_cold/(T_hot−T_cold). Reject to 40 C cooling water and the COP collapses as the cold level drops — so shaft work per unit of cooling *rises* the colder you go.

Cold level needed	Medium	Refrig. T (C)	Ideal COP (reject ~40 C)	Real shaft kW per kW cooling
> 40 C	Cooling water	n/a	—	~0 (pump only)
5-15 C	Chilled water	~5	~6	0.15-0.25 (4-7 kW/kW)
−40 to 0 C	Ammonia / propane	−35	~2.5-3	0.30-0.45 (2-3 kW/kW)
−70 to −40 C	Propylene	−45	~2	0.45-0.55
−100 to −70 C	Ethylene (cascade)	−90	~1.3	0.6-0.9 (~1.3 kW/kW)
< −100 C	Ethylene/methane cascade	−150	< 1	> 1

Rule of thumb: budget ~**1 kW shaft per 3-5 kW of cooling** at the −40 to 0 C levels, climbing past ~1 kW/kW below −100 C. A single-stage cycle is fine for a ~40-50 C lift; beyond that, **cascade** (ethylene rejecting into propylene rejecting into CW) because one compressor over a 5:1+ pressure ratio runs hot and inefficient. In Choupo the refrigeration compressor is just a `compressor` (W_shaft + eta) feeding the cold side of an exchanger; the warm reject side is `coolingWater`, so the power penalty appears explicitly in the work ledger — model it, don't assume it.

When to refrigerate (not before): a low-temperature condenser to recover a light product (C3/C2 overhead), a cold separator to hit a vapour-loss spec, or a reactor that must run cold for selectivity. Always ask first whether **higher pressure** buys the same condensation at a CW level — compressing the overhead is often cheaper than chilling it.

17.2 Vacuum — match the pump to the absolute pressure

Vacuum distillation earns its keep when the bottoms would **thermally crack or polymerise** at the atmospheric boiling point (heat-sensitive: glycols, fatty acids, monomers, vacuum gas oil). Dropping to **0.05-0.1 bar (40-75 mmHg)** can cut boiling points by 80-150 C, keeping the reboiler below the degradation limit and off the high-T utilities (Dowtherm A / fired heaters). Below ~0.1 bar the **column diameter balloons** (vapour density falls, so allowable mass velocity drops); use low-pressure-drop structured packing, not trays.

Absolute pressure	Technology	Notes / WHY
1-0.1 bar	Liquid-ring pump	Rugged, tolerates condensables; service water sets the floor (~30-50 mmHg)
0.1-0.01 bar	Single-stage steam ejector	No moving parts, uses MP/HP steam; cheap but steam-hungry
< 0.01 bar (to ~1 mmHg)	2-3 stage ejectors w/ intercondensers	Stages compound the ratio; condensers cut steam load
Clean, deep, dry	Dry screw pump	Electric, no steam/water effluent; higher capex, watch for fouling

Size ejector **motive steam by the air/leakage load**, not the process vapour — a few kg/h of inleak through flanges sets the bill, so a tight column beats a big ejector.

18 Storage & intermediate tanks

Storage is inventory bought with steel: too little and a late truck or a tripped upstream unit starves the plant; too much and you pay for tankage, land, and a larger fire/spill exposure. Size by **days of stock** tied to the *logistics* of each stream, then pick a roof that matches the volatility. Run the hypothesis in Choupo by feeding the tank's design flow into a flash drum at storage T,P and watching the vapour fraction — if `vf > 0` you have a breathing/venting problem a cone roof cannot hold.

Sizing by days of stock (the WHY = decoupling time-scale):

- **Raw materials: ~7-30 days**. The driver is delivery reliability and shipment size — rail/marine (large parcels, long lead) push to 20-30 days; reliable pipeline or daily truck supply drops to 3-7. Hold enough that a missed delivery does not stop the reactor.
- **Products: ~7-30 days**, set by shipping cadence and demand swing. A monthly tanker means ~30 days plus working heel; spot-truck sales need only a few days.
- **Intermediate / surge: hours to ~2 days**. Purpose is to *decouple* two units so a transient in one does not propagate. Rule of thumb: **5-15 min** of holdup for pump-suction/control surge, **0.5-2 days** between batch and continuous sections (e.g. a reactor feeding a continuous still). More surge = looser control coupling but slower composition response and more off-spec inventory if upstream drifts.

Tank **working volume** = days × daily throughput, then divide by the **fill fraction**: never design to the brim. **Fill ~85-90%**, leaving **10-15% ullage** for thermal expansion (a 30 °C swing on a hydrocarbon adds ~3-4 vol%), foam, and high-level-alarm margin. Overfill is a top cause of major releases (Buncefield), so the ullage is a safety budget, not slack.

Type selection (volatility & pressure drive the roof):

Tank type	Best for	Pressure / vapour-pressure range	Typical size	Why
Cone (fixed) roof	Non-volatile liquids: water, heavy oils, glycols, most products with RVP < ~10 kPa	Atmospheric, near-zero gauge	up to ~10,000 m^3	Cheapest; vapour space breathes in/out — fine when there's little vapour to lose
External floating roof	Volatile liquids: gasoline, naphtha, light solvents (RVP ~10-80 kPa)	Atmospheric	~1,000-100,000 m^3	Roof rides on the liquid → **no vapour space**, cuts breathing/standing losses by ~95% vs cone roof; emissions + fire-risk reduction
Internal floating roof	Volatiles needing weather/rain protection	Atmospheric	medium	Floating deck under a fixed roof — losses control plus debris exclusion
Sphere (Hortonsphere)	LPG, propane, butane, light ends	~2-18 barg (vapour pressure > atm)	~1,000-10,000 m^3	Sphere carries pressure with minimum steel/area; even stress distribution
Bullet (horizontal cylinder)	Smaller pressurised stock: LPG, NH_3, light ends	~5-25 barg	< ~500 m^3	Cheaper than a sphere below ~200-500 m^3; easy to transport and mound

The crossover rule: if the stored fluid's true vapour pressure exceeds atmospheric at the *hottest* expected ambient (summer-skin ~45 °C), it **cannot** sit in an atmospheric tank — it must go pressurised (sphere/bullet). Between non-volatile and pressurised lives the floating-roof band, chosen to cut working+breathing losses you would otherwise vent or send to a flare/recovery unit. Always state the range and let throughput, RVP, and emissions cost decide — there is no single right tank.

19 Process control & instrumentation heuristics

A flowsheet that balances on paper still must hold steady against disturbances. The Pareto question is not tuning — it is **what to pair with what**.

What Choupo can and cannot check for you, stated before you rely on it. An earlier version of this paragraph invited you to "step a disturbance in `ctrl01` and watch whether your loop rejects it or fights another". That invitation was false three ways, and it is corrected here rather than quietly deleted, because a manual that has promised something owes the reader the correction: `ctrl01` carries no disturbance at all (it is a setpoint-tracking case — the disturbance cases are `ctrl02`, `ctrl05`, `ctrl09`); *no* case in the corpus carries more than one PID loop, so "fights another" cannot be watched anywhere; and this section is about *column* pairing, while the dynamic tier has no column in it. A pairing for a column cannot be tested dynamically in Choupo today.

What you *can* do, and it answers the sharpest question in this section — *which tray carries the temperature measurement* — is read it off the rigorous column you already ran. The steady profile gives the slope criterion directly, and re-solving with a manipulated variable nudged gives the sensitivity criterion. The EduTools *Column control* tool does exactly that (its derivation is the Theory Guide's section *Choosing a column control structure, and where the instruments go*) and reports something a single answer would hide: on a 15-stage column the slope criterion picks stage 10, the reflux handle picks stage 11, and the distillate handle picks stage 5 — a different section of the column, with a response fifteen times stronger. The same run shows the reflux response *changing sign* between stages 6 and 7, which no slope of the profile can reveal.

Choupo computes no relative-gain array, at any frequency: there is no steady-state gain matrix in the engine. The pairing heuristics below argue in terms of relative gains, and that argument is the literature's, not a number this simulator produced.

Inventory first. Control every inventory (level, pressure) before any quality — an unbounded inventory shuts the plant down.

Controlled variable	Manipulate	Why
Level (drum, sump, tank)	the OUTflow	self-regulating; never control level by an inflow that sets production
Pressure (vapour)	vent / vapour outflow / condenser duty	fast; pins the vapour inventory
Feed flow / throughput	the feed valve directly	the master production handle
Reactor / column temperature	coolant or steam duty	the quality proxy
Column composition	a sensitive tray TEMPERATURE	direct composition is slow/expensive; tray T tracks it

Tuning starting points. Process PID is **mostly PI** — derivative amplifies noise, so add D only on slow, lag-dominated (temperature) loops. Flow loops: fast and tight. Level loops: loose/averaging (the tank is a buffer). Use the open-loop step (gain, dead time θ, time constant τ) with an IMC/λ rule rather than trial and error.

Structure beyond single loops.

- **Cascade** when a fast inner disturbance exists (jacket-T inside reactor-T rejects coolant upsets early).
- **Feedforward** when a disturbance is MEASURED (feed rate/composition) — act before the error appears, trim with feedback.
- **Ratio** to hold two streams in proportion (stoichiometry, reflux, fuel/air).
- **Override / constraint** to ride a limit (max T, min level) safely.

Pairing discipline: pair each manipulated variable with the controlled variable it affects most directly and with least interaction (a near-1 relative gain). Two loops that fight oscillate — the glass-box dynamic run shows it at once. Keep DOF honest: one independent manipulated variable per controlled variable, no more.

Equipment-specific schemes. Beyond single loops, each unit has a conventional strategy.

Distillation is the hard one: a column has **five** handles (condenser duty, distillate D, reflux L, boilup V, bottoms B) and **five** things to hold (pressure, reflux-drum level, sump level, a top and a bottom composition). Pressure is fixed first (condenser duty or a vent). Each level MUST then be held by an OUTFLOW — and the rule is **control a level with the LARGER stream leaving that drum**: if the reflux ratio R = L/D is high, reflux is the big stream, so hold reflux-drum level with reflux and set D for composition (the "LV" vs "DV" choice). Composition is inferred from a **sensitive tray TEMPERATURE** (fast, cheap) instead of a slow analyser. **Single-end** control

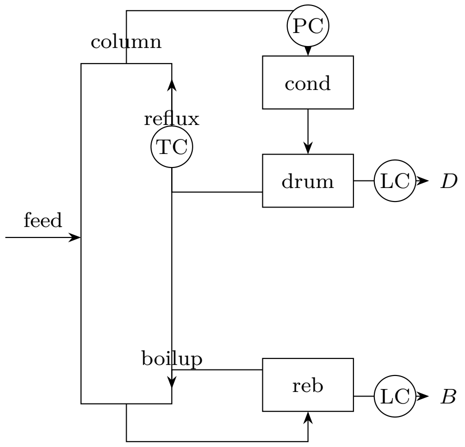


Figure 3: A conventional distillation control scheme: pressure (PC) on the condenser, each level (LC) held by an outflow, composition by a sensitive tray temperature (TC). Five handles, five controlled variables – the pairing is what makes it subtle.

(hold one composition, let the other float) is the robust default; **dual-composition** control saves energy but the two loops interact strongly (high relative gain) and usually need decoupling — which is exactly why students find column control hard. *Reactor:* temperature by coolant/steam duty, usually **cascade** (jacket-T inner, reactor-T outer). *Compressor:* an **anti-surge** recycle loop keeps flow above the surge line. *Fired heater:* outlet T trims the fuel, with an air/fuel **ratio** loop and a **low-flow trip** to protect the tubes.

20 Pressure relief, flare & safety basics

This is a Pareto orientation, not a safety course. The governing standards are **API STD 520/521/526** (sizing, scenarios, devices) and **API STD 537/NFPA 30**; treat the numbers below as where-to-start hypotheses, then run the controlling case.

When a device is mandatory. *Every* closed volume that can be isolated and over-pressured needs an independent overpressure-protection device — no exceptions for "small" vessels. The trigger is the credible-cause inventory, not size. A relief device is the last line of defence: it must work with zero operator action and zero instrument power, which is why a spring-loaded PSV (or rupture disc) is preferred over an instrumented trip for the primary case.

Set vs design pressure (the WHY). Design pressure is set ABOVE the maximum operating pressure so the PSV does not weep on normal swings. Heuristic: **design P = max(1.1 × operating P, operating P + 1.7 bar)**; the two rules cross at 17 barg (0.1 × 17 bar = 1.7 bar): below that the +1.7 bar floor dominates, above it the +10% rule does. The heuristic gives a *preliminary* design pressure, not a MAWP: the **maximum allowable working pressure (MAWP)** is a property of the built vessel — its actual wall, material, temperature and the code it is stamped to — and is at least the design pressure. A single PSV is set at or below the MAWP; at the preliminary stage the set pressure is *assumed* at the design pressure, the lowest MAWP the vessel can have, which is the conservative choice for the relief area. State every pressure as gauge or absolute: the accumulation percentages below apply to the gauge set pressure. Code allows accumulation above set — typically **+10%** for a single non-fire device, **+16%** for multiple devices, and **+21%** for the fire case — because momentary overshoot during full relief is tolerable but permanent yield is not.

Sizing scenarios — size for the worst credible single cause (API 521).

Scenario	Typical controlling load	Why / note
External fire	Heat input to the wetted area, **Q = C · F · A^0.82** (Q in kW for A in m^2; C ≈ 43.2 with adequate drainage and fire-fighting, higher without; a total duty, not a flux)	Often controls vessels with large liquid inventory; sets a large orifice
Blocked outlet	Full pump/compressor/feed flow at shut-in	Common control case for pumped/compressed systems
Thermal expansion	Liquid trapped between two valves, heated	Tiny relief (¾"×1") — but mandatory; cheap to forget
Tube rupture (HX)	High-P side leaks to low-P side; full ΔP flow	The "2/3 rule": if low-side design P < 2/3 high-side, must relieve
Control-valve failure / loss of cooling	Failed-open bypass, reflux/condenser loss	Check fail-safe action of each valve

Size each, then **the largest required orifice wins**; pick the next standard API-526 letter orifice (D through T).

PSV vs rupture disc.

Device	Use when	Trade-off
Spring PSV	Reclosing wanted; clean service	Reseals, testable; can weep, can plug/foul
Rupture disc	Fast/large relief, fouling/corrosive, polymerising	Non-reclosing (full loss of inventory); cheap
Disc + PSV in series	Corrosive service needing reseal	Disc isolates/protects the PSV; best of both

Disposal — flare vs vent. Vent to atmosphere only if the relieved fluid is **non-toxic, non-flammable** at grade and the discharge disperses safely (e.g. steam, air, inert). Flammable/toxic streams route to a **closed flare header → knockout drum → flare**, sized so flare-tip exit velocity stays below ~**0.5 Mach** and grade-level radiation stays under **~6.3 kW/m^2** at the fence (API 521). Choupo **sizes a relief valve but does not model relief**: the sizing pass's `reliefValve` item sizes one valve per protected item on its declared blocked-outlet and fire scenarios, by the same equations as the EduTool *Designing for overpressure* (worked case: `greenAmmonia03_basicDesign`), but no unit relieves a stream, and no PSV-vent enthalpy or flare duty appears in any ledger.

Spacing/segregation (Pareto). Fired heaters and flares are ignition sources — keep ~**15 m** from process units handling flammables; locate control rooms and the flare upwind/away. That figure is an order-of-magnitude orientation, not a value Choupo carries: the binding distances come from the spacing tables and standards the project adopts (insurance guidance such as IRI/GAP, API 521 §5 for flares — not read here), refined later by a dispersion/QRA study. Choupo ships no spacing table; a case's `system/layoutDict` declares its own distances, each with its source and reason, and the engine announces every pair closer than declared (EduTool *Plant layout*, witness `ammonia02_full_plant`). Siting, wind, hazardous areas and the rest of the plot plan are §21.

Governing standards — go to the source; lives depend on it. This section is an orientation only; the binding requirements live in, among others: pressure relief — **API STD 520 & 521**, tank venting — **API STD 2000**, overpressure protection — **ASME BPVC Section VIII** (UG-125+) and **Section I** (boilers); relief-device specs — **API STD 526**; flares — **API STD 521 & 537**; process-safety management — **OSHA 29 CFR 1910.119** and the **AIChE/CCPS** guideline series; functional safety / SIS — **IEC 61508 & 61511**; hazard-study method — **IEC 61882 (HAZOP)**; flammable liquids — **NFPA 30**; electrical area classification — **NFPA 70 / IEC 60079**; facility siting — **API RP 752/753**. Size and select to the standard, then have it independently reviewed. See the Disclaimer & scope at the front of this guide.

21 Plant layout

A flowsheet says what is connected to what; a layout says *where each item stands*, and that second question carries most of the safety, a good part of the piping cost and nearly all of the ease of operation and maintenance. It is also where several numbers the flowsheet never asks for are fixed: how high a column stands on its skirt, how far above its pump a reflux drum must sit, how long every pipe is. This section is a Pareto orientation for a student placing a first plant on a plot — the order in which to decide, and the *why* behind each rule — not a siting study. Every distance that touches safety comes from the standard or the risk study a real project adopts (named below, and see *Disclaimer & scope*); this guide transcribes none of their tables, and neither does Choupo.

The order of decisions. Site and wind first (where a release may drift), then the blocks (process, storage, utilities, loading, buildings) and the roads between them, then the pipe rack that ties the process block together, then the equipment along the rack, and only then the elevations inside each group — columns, drums, pumps — which the NPSH of the pumps decides. A late decision cannot repair an early one: no elevation trick fixes a control room placed downwind of the tank farm.

21.1 Siting and the prevailing wind

- **Read the wind rose.** A release drifts *downwind*, and an ignition source the cloud reaches turns a leak into a fire or an explosion. So the **ignition sources** — fired heaters, boilers, the flare, a workshop, any electrical equipment not certified for a hazardous area — go **upwind or crosswind** of the likely **release points** (pump seals, compressor packings, flanged equipment, sample points, loading arms, relief discharges). Lay out for the wind that blows most often, then check the others: a layout that is safe for one direction only is a bet on the weather.
- **Occupied buildings** — control room, offices, canteen, laboratory, workshop — stand **upwind of the process and the storage**, at the edge of the plant and **near the main gate**, so visitors, contractors and the canteen's deliveries never cross a process area. A control room that must stand close to the units it runs is designed for the blast and toxic exposure of that position (the question API RP 752 and 753 answer), not merely moved a few metres.
- **The flare is two things at once**: an ignition source (keep the flammable clouds from drifting onto it) and a source of radiation, noise and combustion products (keep it away from people). It goes at the edge of the site with its own sterile radius, set by the radiation at grade (§20; the EduTool *Pool-fire radiation and the distance it sets* shows how a radiation threshold becomes a distance).
- **Topography and drainage.** Heavy vapours — LPG, chlorine, a cold ammonia cloud — flow downhill and pool in hollows: keep ignition sources and occupied buildings off low ground, and grade the process areas so that a spill drains *away* from the equipment towards an impounding basin, never under the next unit.

21.2 Separation distances and hazardous areas

Why a distance. A separation does four jobs: it stops a fire or an explosion in one item from escalating to the next (the domino effect — radiation falls with distance, and the distance to a given blast overpressure grows only with the cube root of the explosive mass); it keeps the people in occupied buildings out of the consequence zone; it leaves room to fight a fire and to maintain the plant; and it keeps a released cloud away from ignition sources. Each job gives its own distance, and the largest governs.

Two ways to get the number. *Prescriptive* tables give a minimum edge-to-edge distance between pairs of equipment classes (fired heater to pump, storage tank to process unit, . . .); they are quick, conservative for ordinary plants and silent about unusual ones. *Risk-based* siting computes the consequences of credible releases (radiation, overpressure, toxic dose) and, in a full study, their frequencies; it can justify a tighter plot or demand a wider one. A student design starts from a prescriptive table and checks its largest hazards by consequence. The published sources include the CCPS *Guidelines for Siting and Layout of Facilities*; Mecklenburgh (ed.), *Process Plant Layout* (IChemE); the insurers' *GAP.2.5.2 Oil and Chemical Plant Layout and Spacing*; NFPA 30 (flammable and combustible liquids, tank spacing included), NFPA 58 (liquefied petroleum gas) and API STD 2510 (LPG installations); and, for occupied buildings, API RP 752 and API RP 753.

Hazardous-area classification answers a different question: not “how far apart” but “where may a flammable atmosphere exist, and what electrical equipment may stand there”. Around each *source of release* the zone system (IEC 60079-10-1; the Energy Institute *Model Code of Safe Practice Part 15*; API RP 505; in Europe the ATEX workplace directive 1999/92/EC) draws envelopes: **Zone 0**, where a flammable atmosphere is present continuously or for long periods; **Zone 1**, where it is likely in normal operation; **Zone 2**, where it is not likely and short-lived if it occurs. The extent of each envelope depends on the grade of release, the release rate and the ventilation — which is why flammable service prefers an open, well-ventilated structure to an enclosed building. Equipment inside a zone must be certified for it; open flames and uncertified equipment stay outside every envelope. (NFPA 70 and NFPA 497 carry the North-American Class/Division counterpart.)

What a Choupo case declares. The spacing and zoning tables above grant no redistribution, so Choupo ships none (the licence policy of this project). A case's `system/layoutDict` declares its own separation table between equipment kinds, each rule with a `source` and a `reason`; its own wind (the direction it blows from, the half-angle of the downwind sector, its reach); and, per release source, its own zone envelopes. Write as the source the table or study you actually read: "author-set assumption" is an honest source, a standard you have not opened is not.

21.3 Blocks, roads, fence and battery limits

- **Blocks.** Group the plant into blocks of similar hazard — process units, tank farm, loading, utilities, buildings — each block bounded by roads. Grouping concentrates the hazardous-area envelopes and the fire protection, and the roads double as fire breaks.
- **Roads.** A **perimeter road** and roads between the blocks let a fire tender reach every block from at least two sides and a mobile crane reach every heavy item; avoid dead ends. Keep heavy traffic — trucks, deliveries — at the perimeter, off the process roads.
- **Gates and trucks.** A main gate with a guardhouse, and a second, emergency exit, so that one blocked road does not trap everybody inside. Where raw materials or products move by road, the **weighbridge** and the loading bays sit close to the gate, so a truck is weighed in, loaded and weighed out without entering the process area. A plant whose feeds and products move by pipeline needs neither.
- **Fence and battery limits.** The fence is the security and legal boundary of the site; the **battery limit** is where the plant's scope ends — every feed, product and utility crosses it at a defined point with an isolation valve, a blind and usually a meter. Name what is on the other side (the adjacent park's pipeline, the site cooling-water return): a battery limit nobody labelled is a scope nobody owns.

21.4 Trucks, cars and the people on site

A plant whose feeds and products move by pipe still receives trucks: catalyst at every recharge, chemicals and lubricants, spares, a turnaround's contractors and their cranes, and the waste it sends away. And every shift brings people in cars. Lay out for both before the process block is final, because both need ground and both must stay out of the process.

- **A one-way truck loop.** The truck comes in at the gate, is weighed, drives to where it unloads, and leaves — without reversing and without turning round. A loop driven one way does that with no manoeuvre at all; a dead end needs a **turning circle** at least as large as the truck turns in, and every reversal near people and equipment is an accident waiting. Make the yard roads one-way and wide enough to pass a truck stopped at a door.
- **The design vehicle.** Every corner on the loop must hold the largest truck that may come. In the European Union the law fixes the envelope: any vehicle combination in motion must turn within a swept circle of outer radius 12.50 m and inner radius 5.30 m (Council Directive 96/53/EC, Annex I, point 1.5). So a corner's pavement must reach at least 12.5 m from its centre on the outside, and two corners need straight road between them — one turn must end before the next begins. A swept-path study (the trailer cutting in on the inside of the turn) is the next step a real project takes.
- **The weighbridge** sits on the loop near the gate, on a straight stretch so the truck stops square on the deck, and is passed by every truck the site weighs: weighed in, what a supplier delivers is checked; a site that also ships by road weighs out as well, and needs a second deck on the outbound leg or a deck both directions cross. Cars should not use it.
- **The car park** goes near the gate and away from the process — the first turn after the gate barrier, or outside the security line altogether — so a private car never drives a process road and never meets the truck loop. Its bays are oriented so that a car leaves *forward*: reverse-in bays (the manoeuvre is made on arrival, with time) or drive-through bays. In an evacuation every car leaves at once, and a car park where all of them reverse out blind is a hazard of its own. A car is also an ignition source: keep the car park out of every hazardous area and out of the downwind sector of the release points.
- **Offices and the visitors' entrance** stand by the main gate, upwind and outside every separation from the process, so a visitor or a supplier reaches them without crossing the plant.
- **Two ways in.** A second gate on another side, normally locked, is the fire brigade's way in when the main gate is blocked — by the fire, by a tanker, by the evacuation leaving.
- **Muster points** — where people gather and are counted — are on the upwind side and away from every release point, and there are at least two, on opposite sides of the site, because the wind turns: when it puts one in the cloud, the other is clear. A **windsock**, in sight of the control room and of the muster points, tells people which way to walk.
- **Fire water.** The fire-water reserve and its pumps stand where a fire in the plant cannot reach them — upwind and well separated from the process — because the water is needed most when the process is burning. The ring main and its hydrants follow the roads round every block.
- **A laydown area** — hard standing beside the process block and on the truck loop — takes the catalyst bins, the exchanger bundles, the scaffolding and a crane's outriggers during a turnaround.

21.5 Pipe racks

Most of a process block's piping runs on one **main rack** — the plant's spine — with the equipment lined up on **both sides** of it. The rule that follows is simple: items joined by large or expensive lines stand *close* and in *flow order* along the rack, so most lines run straight along it.

- **Pumps** stand in a row under or just beside the rack edge, their suction vessels on the outer side, their motors towards an access way.
- **Air coolers** usually sit **on top of the rack**: they need air, not plot area, and their condensate drains down to drums at the rack side.
- **Tiers.** Lines running one way on one tier and the other way on another, so that a change of direction is a change of elevation and no line clashes with another; hot and large lines on the side nearest the columns, utility headers on their own tier; spare width for the lines a later revamp will need.
- **Headroom.** Wherever the rack (or any pipe) crosses a road it must clear the largest vehicle that uses that road — a fire tender, a mobile crane — by the owner's standard. Where it cannot, the road moves, not the rule.

21.6 Tank farms and the choice of storage

Storage holds the largest inventories on the site, so the tank farm goes at the edge of the plot, away from the process and the occupied buildings, on lower ground and downwind where the site allows, and close to the loading area. The tank type follows from the vapour pressure at the hottest storage temperature (the crossover rule of §18); a liquefied gas has three answers:

Storage	When	Why (glass-box)
Pressurised at ambient temperature (sphere, bullet, mounded bullet)	small to medium inventories	no refrigeration plant; the wall carries the vapour pressure at the hottest ambient, so the steel per tonne stored stays high however large the store
Semi-refrigerated (moderate pressure, below ambient)	medium inventories	a modest refrigeration duty buys a lower design pressure and a thinner wall
Fully refrigerated at about atmospheric pressure, at the normal boiling point	large inventories	a thin-walled, insulated tank of almost any size; the boil-off is recompressed and recondensed by a refrigeration unit that runs for the life of the tank

Ammonia is the textbook case. Its vapour pressure is about 8.6 bar at 20 °C and close to 18 bar at 45 °C, so storage at ambient temperature needs pressure vessels designed for the summer. A few hundred tonnes sit comfortably in bullets or a sphere; tens of thousands of tonnes would need a field of spheres, so large inventories are stored **refrigerated at about −33 °C**, the normal boiling point, in insulated atmospheric tanks with a refrigeration unit for the boil-off. The crossover is economic, not physical, and is usually placed in the low thousands of tonnes, where either answer can be argued. Choupo carries both: `greenAmmoniaPEQ_baseCase` stores its product refrigerated at atmospheric pressure, and `ammonia02_full_plant` keeps two days of product as a pressurised liquid in bullets.

Why gas is rarely stored. Compressed hydrogen at 200 bar and 25 °C holds about 15 kg/m^3; liquid ammonia about 600–680 kg/m^3. And a gas store gives back only the gas between its full and its emptied pressure, so a gas is *buffered* for hours, not stored for weeks — which is why plants move gases by pipeline and store liquids. The hydrogen buffer of `greenAmmoniaPEQ_baseCase`, between an intermittent electrolyser and a steady synthesis loop, is the exception that shows the rule.

Containment. Atmospheric tanks of flammable or toxic liquid stand in a **bund** (dike) that holds at least the largest tank's contents, with the margin the applicable code requires; group compatible products in one bund, leave fire-fighting access round it, and route no process pipe through it. Refrigerated tanks of liquefied gas are often double- or full-containment, the outer wall acting as the bund. Orient bullets so their ends do not point at occupied buildings or other vessels: a bullet that fails in a fire travels along its axis. Detailed tank spacing is again a code question (NFPA 30, NFPA 58, API STD 2510; API STD 620 for low-pressure and refrigerated tanks).

21.7 Secondary containment: bunds and impounding areas

Every vessel that holds a liquid will one day leak: a flange, a drain left open, a tank overfilled, a shell that fails. **Secondary containment** decides where that liquid goes. It is the layout decision that limits three things at once:

how far the liquid spreads (into the drains, the soil, the river), how large a pool evaporates (the size of the toxic or flammable cloud follows the pool's surface), and how large a fire can burn (a pool fire's radiation scales with the pool, the EduTool *Pool-fire radiation and the distance it sets*).

- **A bund** (dike, retention basin) is a wall round the tanks, on an impermeable floor: the spill is held *where it falls*. Its **net capacity** is its floor area times its wall height, less the volume the tanks and their foundations occupy below the wall's top — a low wall round tall tanks loses a large share of its volume to them. The usual form of the capacity rule is a margin over the **largest** tank (the largest tank fails, not all of them at once), sometimes with a second term on the **total** that governs when it is larger, plus an **allowance** for the rain the bund collects and the fire water poured into it during a fire. Which margin, which allowance and which design rainfall is the applicable code's or guideline's: read it and cite it.
- **Low or high wall.** A low wall spreads the pool over a large floor (more evaporation, a larger fire, but fire-fighting reaches over it); a high wall keeps the pool small and under the tanks, and makes access and escape harder. Ramps or stairs over the wall, never a gap in it; pipes carried *over* the wall, never through it (a penetration the spill can leave by).
- **The drain.** Rain collects in every bund; it leaves through a valve that is kept **closed** and opened by hand only when the water is seen to be clean, or by a sump pump. A drain left open to the sewer empties the bund on the day it is needed.
- **Impounding** (a remote catch basin) is the other answer: a kerbed, sloped catch area under the vessels leads the spill *away* to a pit at a distance, so the pool — and any fire on it — forms clear of the vessels, and its small surface limits evaporation. It suits liquefied gases stored under pressure, where a spill flashes in part and the rest pools cold: the liquid must not be allowed to lie under the bullets. Refrigerated tanks are often double- or full-containment, the outer wall acting as the bund.
- **Compatibility and grouping.** Liquids that react together never share a bund; a large tank farm is divided by intermediate walls so that one spill does not cover every tank.

In a Choupo case. A bund or an impounding area is a *declared region* in `system/layoutDict` (`bunds ({ name; kind bund|impounding; items (...); region { ... } or polygon (...); wall { height; } [pit { x; y; length; width; depth; }] capacityRule { basis vesselVolume|heldLiquid; terms ({ fraction; of largest|total; } ...); [allowances (...);] source; reason; } drain; source; reason; });`), not a placed item, so the overlap rule does not apply to it. The engine (`src/postProcessing/layout/PlantLayout.cpp`, `readBunds`) **refuses** a wall that cannot be built — a listed item outside it, an unlisted item inside it, a wall through a footprint, a pit under an item or a road, two containments sharing a floor, one across the fence — and a containment that declares no capacity rule: Choupo ships no code's table, so the rule, its terms and its source are the case's. It computes the net volume (the displaced volume of a vertical cylinder, a lying cylinder's circular segment, a box), the required volume (the largest of the declared terms on the declared basis, plus the allowances; a `depth` allowance is a rainfall over the catch area and the pit) and publishes the verdict. It **announces** a containment short of its rule, a drain open to the sewer, an impounding pit inside its own catch area, a road through a wall, a pipe through one below its top — and, on every layout, an item in the open that holds liquid by the run's own inventory and that no containment encloses. The witnesses: `greenAmmoniaPEQ_baseCase` (a 4 m bund round the two refrigerated tanks, and a kerbed slab under the loop's three liquid vessels draining to a remote pit; lower the bund's wall to 2 m and read the shortfall) and `ammonia02_full_plant` (an impounding area under the seven bullets, its pit remote). Not checked: the floor's slope and its impermeability, the wall's strength and thickness, intermediate walls, the evaporation from the pool, and whether a declared margin or rainfall is the right one.

The EduTool *Secondary containment: designing a bund* teaches the same arithmetic — its TypeScript is held to the engine's (`src/postProcessing/layout/BundArithmetic.H`) by `check_bund_arithmetic` — and goes beyond it on the page: the useful capacity in which the failing tank's own footprint is free (the engine deducts every footprint, the safe side), a sloped floor, a freeboard, the level a spill reaches, and a rupture, a leak, rain and an open drain integrated in time until the bund overtops.

21.8 The distillation column group

A column is not one item but a **group**: the shell, its reboiler, the overhead condenser, the reflux drum and two pumps (reflux and bottoms). They are laid out together, beside the rack, and their elevations follow from two rules: liquid must reach every pump with enough NPSH, and vapour must flow without trapping liquid.

The reboiler stands at the base.

- **Vertical thermosiphon** (the common choice for clean service): bolted beside the column, close-coupled; the liquid head between the column sump and the reboiler drives the circulation, so their relative elevation is a design variable, not a detail.
- **Kettle**: a horizontal exchanger at grade or on a low structure, fed from the column bottom by gravity; it is itself an equilibrium stage, and the bottoms product leaves from its overflow compartment — so the *kettle*, not the column, is the bottoms pump's suction vessel, and it is the kettle that must be elevated for NPSH.

- **Forced circulation**: a pump drives the reboiler loop (viscous or fouling bottoms, or too little head for a thermosiphon) — one more pump to place and to give NPSH.

The overhead runs, from the top down: column top → condenser → reflux drum → reflux pump.

- **Pumped reflux** (the common case): the condenser sits on an exchanger structure beside the column, or as an air cooler on top of the rack; the reflux drum stands below it, so the condensate drains by gravity; the reflux pump stands at grade below the drum and sends both the reflux (back up to the top tray) and the distillate. The condenser need *not* be above the column top: the overhead vapour line falls from the column to the condenser.
- **Gravity reflux** (small columns): the condenser is mounted above the column top, on the column or on a structure, and the reflux runs back by gravity through a liquid seal — no pump, but a heavy exchanger high in the air and awkward to maintain, acceptable only when it is small.
- **The overhead vapour line** is short, large and **self-draining**: it slopes continuously towards the condenser, with no low point where condensate can collect and slug. The same holds for every two-phase line in the group.

21.9 NPSH and the elevations it fixes

The NPSH rule of §15.4 is where the layout and the pump meet. The net positive suction head *available* is

$$\mathrm{NPSH}_a \;=\; \frac{P_{\text{surface}} - P_{\text{vap}}}{\rho\, g} \;+\; z \;-\; h_f,$$

where P_{surface} is the pressure on the liquid surface in the suction vessel, P_{vap} the vapour pressure of the liquid at the pumping temperature, ρ its density, g the acceleration of gravity, z the height of the *lowest* operating liquid level above the pump's impeller centreline, and h_f the suction losses (pipe friction, entrance, fittings, strainer) in metres of the liquid pumped. The pump runs without cavitating while NPSH_a exceeds the NPSH_r its maker measured, by a margin (§15.4; ANSI/HI 9.6.1 is the guideline on NPSH margin).

The key observation. A column sump, a kettle and a reflux drum all hold liquid at its **bubble point**: the vapour above it is in equilibrium with it, so $P_{\text{surface}} = P_{\text{vap}}$ and the first term vanishes:

$$\mathrm{NPSH}_a \;=\; z - h_f.$$

Pure geometry: no pressure, however high, and no density enters. The only lever left is *height* — which is why a column stands on a tall skirt and a reflux drum on a structure. Subcooling the reflux adds the first term back ($\Delta P_{\text{vap}} \approx (\mathrm{d}P_{\text{vap}}/\mathrm{d}T)\,\Delta T_{\text{sub}}$), but most designers take no credit for it: the liquid surface in the drum sits under its own vapour and re-saturates.

Worked example (illustrative numbers, not the data of any real pump or column). A bottoms pump has $\mathrm{NPSH}_r = 3.0$ m at its rated flow; take a margin of 1.0 m and suction losses $h_f = 0.7$ m. Then $z \geq 3.0+1.0+0.7 = 4.7$ m. With the impeller centreline 0.6 m above grade, the low liquid level must stand at least $0.6 + 4.7 = 5.3$ m above grade; if that level is 0.4 m above the column's bottom tangent line, the tangent line stands at 4.9 m — a skirt of about 5 m. For the reflux pump, with $\mathrm{NPSH}_r = 2.0$ m, the same 1.0 m margin and $h_f = 0.5$ m, $z \geq 3.5$ m: the drum's low level stands at least 4.0 m above grade (centreline at 0.5 m) and, that level being 0.3 m above the bottom of a horizontal drum, the drum bottom at 3.7 m. The drum therefore sits on the first floor of the exchanger structure and the condenser on the floor above, so the condensate still drains by gravity. Had subcooling been credited, 5 K of it on a liquid with $\mathrm{d}P_{\text{vap}}/\mathrm{d}T \approx 3$ kPa/K and $\rho = 750$ kg/m^3 would have added $15\,000/(750 \times 9.81) \approx 2.0$ m — the size of the whole margin, and the reason a hot pump suction is never designed on a subcooling that may not be there.

Levers when the elevation costs too much: a low-NPSH pump (a larger eye, an inducer, a lower speed), a vertical can pump whose impeller sits below grade, a shorter and fatter suction line (low velocity, an eccentric reducer flat side up, no high point where vapour can gather), or a booster pump. Each lever has a price; the extra metre of skirt is often the cheapest.

21.10 Platforms, ladders and walkways

- **Each tall column is self-supporting** on its own foundation, carrying its own wind and seismic loads; it is not hung from a structure. Where it passes through the floors of an adjacent structure, the opening leaves clearance for the different thermal growth of a hot shell and cold steelwork.
- **Platforms** go wherever somebody must stand: at every manhole, at the valves and instruments that are operated or maintained, at the relief valves, and at the top, usually with a davit to lift internals and valves. A column therefore has a **piping side**, facing the rack, where the risers run, and an **access side** with the manholes and platforms.

- **Ladders** are caged above a set height and broken by rest landings at set intervals; both are fixed by the applicable regulation (ISO 14122-4 for fixed ladders in Europe, OSHA 29 CFR 1910 Subpart D in the United States, or the local code) — read it rather than guess the number. A long climb is often better served by a stair tower.
- **Walkways** join adjacent columns, or a column and the exchanger structure, at common platform elevations. They give every high platform a **second escape route** — a single ladder is a single way down — and let an operator move from one column to the next without descending to grade.

21.11 Maintenance and crane access

Lay out for the day the plant is opened, not only for the day it runs:

- **Tube bundles** are pulled: leave a clear space in front of each shell-and-tube exchanger's channel end at least as long as the bundle, on the road side, with room for the extractor.
- **Column internals and catalyst** come out through the manholes and the top: a mobile crane must reach the column, so keep its access side free, **opposite the rack**, with hard standing for the crane and room to lay down trays, packing or catalyst bins.
- **Pumps and compressors** need room to pull a rotor and a motor, and lifting beams or crane access over the larger machines.
- **Removable spools** and blinds at every exchanger, vessel and pump, so an item can be isolated and taken out without cutting pipe.

21.12 What Choupo checks, and what it does not

A case declares its plot plan in `system/layoutDict` (the grammar is in `docs/ai/case-layout.md`, the design record in `docs/design/a-plant-layout-is-declared-and-checked.md`); the check lives in `src/postProcessing/layout/PlantLayout.cpp`. Read against that code:

Checked.

- Every *sized* item is placed at a declared plan position (x, y) and base elevation z; its footprint is the item's own published diameter and length, or a box the case declares where the sizer published none. Two footprints that **overlap** in plan and in height are **refused**, and so is a tray stack, a catalyst charge or a relief valve placed on its own (the refusal names its host).
- A pair closer than its **declared separation**, an item outside its sector's declared area, and an ignition source inside a declared **zone envelope** or **downwind** of a release source are **announced**: the run continues, because a short distance is a layout the student must defend, while an overlap is not a layout at all.
- Every material stream is **routed** orthogonally over one pipe rack (declared, or a stated default that is announced), or listed as not routed with its reason; each route's length is published.
- **Roads** are checked against the footprints, the rack and the pipes (a road through an item, or under a rack or pipe lower than the declared headroom, is refused), and an item farther than the declared access distance from a road the gate reaches is announced. The **fence** is derived from the plot and opens at the gates; a road that meets it anywhere else is refused, and every boundary stream crosses it at a battery-limit point carrying the label the case declared (an unlabelled crossing is announced).
- The **plot area** is published (grown by a declared margin; absent one, a lower bound, announced), and the economics prices it as land when the case declares a land price.
- **Truck routes** are walked against a declared **design vehicle** (its turning diameter, with its source): a declared route — an ordered list of roads from a gate back to a gate — is followed from one road onto the next where they meet, and the engine announces a reversal with no turning area big enough, a corner whose pavement (the declared corner radius plus half the narrower road) is short of half the turning diameter, two corners closer than their two radii, a one-way road driven backwards, a weighbridge or a declared stop the route misses, and a route that starts or ends anywhere but at a gate. Each road is carried on from its end to its gate's opening in the fence.
- A **car park** declares its bays (size, aisle, angle, arrangement, how a car enters); the engine lays the rows out, refuses a count the footprint cannot hold, and announces bays a car leaves by reversing. A muster point or the fire-water pump house inside a release source's zone or downwind of it is announced, and so is a site with a single entrance.
- **Secondary containment**: a declared **bund** or **impounding area** is held to its own declared capacity rule (net volume against the required one, the verdict published), a wall that cannot be built is refused, and an item in the open that holds liquid with no containment round it is announced (§21.7).
- On storage: the atmospheric tank sizer refuses a gas and a liquid above 1.5 bar, and prices a tank below 0 °C as a lower bound (`src/postProcessing/sizing/StorageTankSize.cpp`); a gas buffer is sized as pressure vessels (`gasStorage`, `src/postProcessing/sizing/GasStoreSize.cpp`) and a liquefied gas as a pressurised liquid store (`liquefiedGasStorage`).

Not checked — yours to do by hand.

- **The vertical order of a column group, and the NPSH of every pump.** The column's shell, condenser, reboiler and reflux drum are separate sized items (`src/postProcessing/sizing/ColumnSize.cpp`), each placed at the z the case declares; nothing relates those elevations to one another or to a pump, and the engine publishes no NPSH of either kind. A condenser placed at grade beside its column is accepted (`aceticAcid02_plant` does exactly that, and its `layoutDict` says so). Do the worked example above for your own pumps.
- **The column's own piping.** A unit that realises several items is piped through the first of them the case places, so the overhead loop and the reboiler circuit are not drawn and their lengths are not counted.
- **The separation and zoning values themselves.** The engine tests the distances and envelopes the case declares; it never says whether they are right. Neither the pressure drop nor the cost of a pipe is computed from its route.
- **Occupied buildings and containment.** A control room, office or canteen is tested only as a declared ignition source (and against the separations the case declares), never for blast or toxic exposure; escape routes, crane and bundle-pulling space are not modelled. Bunds and impounding areas *are* checked (§21.7), but only their volume and their geometry: not the floor's slope, the wall's strength, nor the pool that would evaporate.
- **What a truck sweeps.** The corner check is the outer edge only: the trailer cutting in on the inside of a turn (the swept path), the road's gradient and load class, a truck stopped at a door blocking the loop, and the fire-water ring main and its hydrants are not modelled; the car-park rule lays out straight rows between end aisles and does not check how a car reaches them.
- **Ladders and platforms are drawing, not data.** The 3D view gives every vertical item taller than a fixed height a caged ladder and platforms at a fixed spacing, flagged as decoration (`gui/src/case/equipmentModel.ts`); a column shell is stamped `vessel` like every drum, so the view cannot tell a tower from a tall drum. No platform, ladder or walkway is placed, counted or checked by the engine.

Run it. Open `tutorials/plant/ammonia02_full_plant` (declared rack, roads, two gates, offices, car park, weighbridge and a one-way truck loop, laydown, muster points, fire water, a windsock, the fence and labelled battery limits), move the control room into the downwind sector of a release source, and read the finding; then lower the main road's `cornerRadius` to 5 m and read what the truck route says. Then open `tutorials/plant/aceticAcid02_plant`, lift its column's condenser onto a structure, and work out by hand the skirt and the drum elevation its pumps would need. The EduTool *Plant layout: where each item stands, and why* walks the same ground with guided moves; *Pump vs system curve* treats NPSH on its own.

Further reading. Mecklenburgh (ed.), *Process Plant Layout*, IChemE; CCPS, *Guidelines for Siting and Layout of Facilities*, AIChE; Bausbacher & Hunt, *Process Plant Layout and Piping Design*, Prentice Hall; Kister, *Distillation Operation*, McGraw-Hill (the column group in operation); and, for anything that touches safety, the standards named above.

22 Effluent treatment — liquid & gaseous

Every flowsheet has a back door: the streams you do NOT sell. Design them with the same care as the product train — environmental limits are *legal* limits, and an untreated vent or outfall can shut a plant down. **Hypothesis to run:** before sizing any treatment, MINIMISE the load at source (recycle, substitution, leak reduction) — the cheapest effluent is the one never made.

22.1 Liquid effluent (wastewater)

A train of increasing cost and specificity; stop when you meet the discharge limit (typically COD, BOD, suspended solids, pH, oil & grease, specific toxics):

Stage	Removes	Typical numbers	Use when
Equalisation + neutralisation	surges, pH	discharge pH 6–9; hours of buffer	always first
API separator / DAF	free oil, solids	oil to < 10–50 mg/L	oily / heavy-solids water
Biological (activated sludge)	BOD / COD	85–98% BOD removal; F/M 0.2–0.5	biodegradable organics
Adsorption (GAC) / advanced oxidation	refractory toxics	polishing to ppb	non-biodegradable, colour
Membrane (UF/RO) / evaporation	dissolved salts	water reuse, zero-liquid-discharge	reuse or brine concentrate

WHY a train: each stage is cheap *only* for what it targets — biology cannot take free oil, carbon is wasted on bulk BOD. Match the stage to the contaminant.

22.2 Gaseous effluent (air emissions)

Pollutant	Control	Typical numbers
Particulates	cyclone → bag filter / ESP	baghouse to < 10 mg/m^3; cyclone only > 5–10 µm
Acid gases (SOx, HCl)	wet / dry scrubber	90–99% removal; L/G 5–15 L/m^3
VOC, dilute & high-flow	thermal / regenerative oxidiser	95–99% destruction at 760–820 °C; RTO recovers ~95% heat
VOC, concentrated / valuable	condensation / carbon / absorption	recover the solvent if it pays
NOx	low-NOx burner + SCR	SCR 80–90% at 300–400 °C with ammonia

The master rule: **dilute-and-high-flow favours destruction (oxidiser); concentrated-and-valuable favours recovery (condense/absorb)** — the same see-then-decide trade-off as everywhere else. Choupo's cyclone, absorber, and combustion duties size the front end; the discharge limit is the spec and the local regulation is the law (see *Standards & further reading*).

23 Pharmaceutical facilities — the GMP interior

A chemical plant is laid out on a plot (*Plant layout* above); a pharmaceutical plant is laid out *inside a building*, and the questions change. Where an item stands matters less than which *room* it stands in, how clean that room is held, which way the air moves through its doors, and who and what walks through it. This section is the Pareto orientation for an oral-solid-dose (tablet) facility — the case a student group meets first — and it says what Choupo checks when the case declares the interior, and what it leaves to the student.

Cleanliness grades. EU GMP Annex 1 (European Commission, EudraLex Volume 4, *Manufacture of Sterile Medicinal Products*, C(2022) 5938, para. 4.4) defines four grades: **A**, the critical zone for high-risk aseptic operations; **B**, the background room for grade A; **C** and **D**, cleanrooms for the less critical stages. Annex 1 is written for *sterile* products. A tablet plant is not sterile, and its rooms are usually specified by the company's own risk assessment, often as "controlled, not classified" (CNC) or by analogy with grade D; the grade a case declares is therefore the *case's* statement, with its source and reason, never a fact the software looks up.

The pressure cascade. Air must move from the cleaner room to the less clean one, so the cleaner room is held at the higher pressure. Annex 1 para. 4.14 gives a guidance value: adjacent rooms of different grades differ by at least 10 Pa. Choupo takes the step from the case (`cascade { minimumStep; source; reason; }`) and announces every door across which the cleaner side is not higher by that step.

Airlocks. Annex 1 calls an airlock "an enclosed space with interlocked doors" (glossary) and asks for one wherever people or material move between different grades, separate for personnel and material where possible (para. 4.12). An airlock is a *room* with its own pressure, and its pressure decides what it does:

- a **cascade** airlock sits between its two neighbours, a step in the cascade;
- a **bubble** airlock is higher than both, pushing air out through both doors — it keeps the dust of a contained room in *and* the corridor's air out;
- a **sink** airlock is lower than both, drawing air in through both doors and exhausting it.

A changing (gowning) room is a personnel airlock in stages of increasing cleanliness (para. 4.12 (i)); its final stage, at rest, is of the grade of the room it leads to.

Containment. A potent or toxic powder — the active ingredient while it is dispensed, sprayed or blended — must stay in its room. The room is held *negative* to its neighbours and reached through bubble or sink airlocks, Annex 1 para. 4.14's "positively or negatively pressurized airlocks". This is where the cascade and the containment pull in opposite directions, and the airlock is how a design reconciles them.

Flows. Three flows are laid out and kept apart: *personnel* (street clothes in, gowned to the work room), *material* (raw material in, product out — one way, never back through a room it has left) and *waste* (out by its own route). Where two flows must share a room (one clean corridor in a small plant), the design says so and why (closed containers, time separation by procedure).

Solvents. An amorphous solid dispersion is often made by spray drying from an organic solvent. The suite is then a hazardous area (gas zone 0, 1 or 2; a dust zone 20, 21 or 22 where powder clouds form) and is kept apart from ignition sources (switchgear, non-rated motors) by a separation the project's own classification sets.

Escape. Every room has a way out — doors and stairs, not a lift.

23.1 What Choupo checks, and what it does not

A case declares the interior as `building { ... }` in `system/layoutDict` (grammar: `docs/ai/case-layout.md`; engine: `src/postProcessing/layout/BuildingLayout.cpp`). The engine **refuses** a building that cannot exist: two rooms sharing a volume, a door between rooms with no common wall, an item that does not fit its room, a road through a room. It **announces** — on the console, in the caveat block and in the result — every door that breaks the cascade, every airlock whose pressures do not realise its declared regime, every change of grade with no airlock, a person who enters a graded room un-gowned, a contained room that is not negative or is reached directly, a flow through a wall, a material flow that goes back, two flows crossing where nobody declared they may, an ATEX room too close to an ignition room, and a room with no way out. The GUI draws the building in cutaway, coloured by grade, with the flows as arrows. Each of these rules is taught as an exercise, one changed declaration at a time, on a small teaching building the engine checks in the browser (EduTool *The GMP interior of a tablet plant*, beside *Plant layout*); the corpus witness with a whole building declared is `apalutamide01_tablets`.

Air and water, sized from the building. A case's `cleanUtilities` pass reads the rooms the layout has just checked (plan area, floor height, grade) and sizes the air plant from *air-change rates the case declares, one per grade, each with its source and reason*: supply flow, fan power and the sensible heating and cooling of the fresh-air share. The air is sensible only — no humidity, no latent load, no room heat gain, and the run says so — and the purified-water generator and storage follow from the water the batch plan draws and the cleans the plan counts (below). No table of air changes is shipped (the published ones are standards that grant no redistribution), so the rates in the two tablet witnesses are author-set and marked so.

The air has a price. Design reports typically carry the HVAC as one lump; here its capital follows from the supply air just computed. The case *declares an installed cost per m^3/h of supply air for each grade* (a `costing { quotes }` entry `hvacGrade`, `valuePerSupplyAir`, with its currency, year, source and reason), and the costing multiplies it by the supply air the pass sized for that grade: the HVAC capital is published per grade and in total. A grade with no declared cost is refused by name and the capital total reads incomplete — nothing is defaulted, and the two witnesses' figures are author-set placeholders, to be replaced by a quotation. The building quote must not carry the HVAC as well.

Not checked — yours to do. The air-change rate each grade needs, HVAC zoning and the filter classes behind each grade; particle and microbial limits; door widths, corridor widths and travel distances to an exit; the interlocks themselves; and whether a declared grade, step or separation is right for the product — that is the project's risk assessment.

Further reading. European Commission, *EudraLex Volume 4, Annex 1: Manufacture of Sterile Medicinal Products* (2022), whose reuse the Commission's document-reuse policy allows with acknowledgement of the source; ISPE, *Baseline Guide Volume 2: Oral Solid Dosage Forms*; ISPE, *Baseline Guide Volume 7: Risk-Based Manufacture of Pharmaceutical Products*; WHO, *Good manufacturing practices: guidelines on heating, ventilation and air-conditioning systems for non-sterile pharmaceutical products.* The ISPE guides are cited by title only: their tables grant no redistribution and none is transcribed here.

23.2 Batch and campaign operation — sizing per lot

A drug-product plant does not run at a steady rate. Its steps are done in **lots** — a blend of a few hundred kilograms, released by quality control before the next step uses it — and its continuous machines (a closed-loop spray dryer) in **campaigns** of several lots between full cleans. Sizing a blender for the plant's *average* hourly flow designs a machine no lot ever fills; it must be sized for one lot.

The classical method for a single-product batch plant works on the material balance *per kilogram of product*, which a steady flowsheet gives exactly. For each step (*stage*) j:

- the mass it handles per kg of product, f_j, from the balance; the **size factor** of a vessel is $S_j = f_j/(\rho_j \varphi_j)$ (bulk density, fill fraction), and the vessel for a lot of B kg of product holds $V_j = S_j B$; a semicontinuous machine (press, compactor, packaging line) is sized by its rate, $f_j B$ over its operating time;
- its **cycle** t_j is charge + process + discharge + clean; a vessel that empties through the step it feeds is held for that step's whole run;
- the **limiting cycle time** of a train is $T_L = \max_j t_j/N_{op,j}$ — the slowest step paces every lot, and a second unit *out of phase* (taking alternate lots) halves its share, while a unit *in phase* (sharing a lot) halves its size;
- an **intermediate store** decouples two trains, so each keeps its own cycle, campaigns and shifts (the spray dryer round the clock, the tablet suite on two shifts);
- lots a year are $N = \lceil Q/B \rceil$; a campaign takes the first lot's pass through every step, then one lot every T_L, then the changeover; a step's **occupancy** is its busy hours over its train's available hours.

A step that never limits and is busy a small fraction of the cycle is oversized or could be shared (one blender for both blends); a step that would be busy longer than the year cannot run the plan.

In Choupo a `batchPlan {}` pass in `system/postDict` does this on the converged flowsheet (grammar: `docs/ai/case-layout.md`; method: `src/postProcessing/BatchPlanPass.H`): it reads every stage's mass from the run, takes every time, density, fill, lot and shift pattern from the case with its source, refuses an impossible plan and announces a poor one; a vessel may take its volume from it, a quoted machine's price may scale with its size (the cost-capacity exponent is the case's), and the economics takes the year's flows from its production. **Cleaning, counted by the plan.** A plan may declare, for each wet-cleaned machine, the purified water of *one clean of one instance* (after a lot, and at a campaign changeover) and the fraction of that water that leaves as effluent (`cleaning`, each with its source and reason). The plan itself counts the cleans — one after every lot of each instance the lot is split over, and one of every installed instance at each changeover — and publishes, per machine, per full campaign of each train and per year, the cleans, the purified water and the *volume* of effluent (m^3 a year; its load is not computed). The purified-water generator and tank of `cleanUtilities` draw these cleans in the plan's Gantt: there is one water ledger, so a lump "cleaning" use beside them would count it twice. Without a `cleaning` declaration nothing is counted, and the run says so. A `ratedMachine` entry then asks, of each machine the flowsheet carries as a mass balance only, *does it fit the lot and the shift*: the load (working volume, kg/h, tablets/h or packs/h) read from the plan and the run, over a vendor rating the case declares with its source — never defaulted — and refused above 100 %, announced above 85 %. The app's Reports tab draws it: a machine-utilisation table, the bottleneck first, with the engine's own utilisation, its "announced" mark and the batch plan's stage utilisation beside it, and the Gantt chart of the first campaign (SVG download); nothing in it is recomputed by the app. It computes no machine physics (no compaction force, no blend uniformity). **Not computed:** the intermediate store's size, a second product sharing the line, QC release times. Worked case: `tutorials/batch/plant/apalutamide01_tablets`. The method is taught in the batch-process chapters of design textbooks (Biegler, Grossmann & Westerberg, *Systematic Methods of Chemical Process Design*; Seider et al., *Product and Process Design Principles*) — cited by title only.

24 Cost estimation & profitability

Costing in Choupo is a *post-processing pass* (`CostingPass`, Guthrie/Turton models) feeding `EconomicsPass` (see `optim05_reactor_npv`). The numbers below are heuristics to be *run and checked*, never final verdicts — accuracy depends entirely on the AACE estimate class.

Capital (CAPEX). Two routes, increasing in effort:

- *Lang factor* — fixed capital = Lang factor × Σ(delivered equipment cost). Hypothesis to verify: a fluids plant lands near 4.7×; solids-handling near 3.1×; mixed 3.6×. The factor absorbs piping, instrumentation, civil, and indirects, so it is crude (±30-40%) but fast.
- *Bare-module (Guthrie/Turton)* — per-unit: $C_{BM} = C_p^0\, F_{BM}$, with $F_{BM} = (B_1 + B_2 F_M F_P)$. Pressure ($F_P$) and material ($F_M$) multipliers are *itemised*, so this is the glass-box default. Verify: a SS316 vessel at 30 bar shows $F_{BM} \approx 5 - 7$ vs ~3 for carbon steel at 1 bar.

Item	Typical range	Why
Lang factor (fluids)	4.5-5.0	Indirects dominate fluid plants
Lang factor (solids)	3.0-3.6	Less piping/instrumentation
Bare-module F_{BM} (CS, low-P)	2.5-4.0	Base installation overhead
Six-tenths exponent n	0.5-0.7 (≈0.6)	Economy of scale on area/volume
CEPCI escalation	index ratio	~600 (2019)→~800 (2023)
Site/auxiliary uplift	+10-25%	Grassroots vs battery-limit

Six-tenths rule: $C_2 = C_1(S_2/S_1)^{0.6}$. Doubling capacity raises cost only ~52% (2ˆ0.6) — verify before sizing a redundant train. *CEPCI:* $C_{now} = C_{base}\,(I_{now}/I_{base})$; skipping a 4-year escalation understates CAPEX ~25%.

Operating cost (COM). Rule of thumb (Turton): $COM \approx 0.18\,FCI + 2.73\,C_{OL} + 1.23(C_{UT} + C_{WT} + C_{RM})$. Verify the dominant term:

COM component	Typical share	Why
Raw materials	40-80%	Usually dominant; selectivity is king
Utilities	10-30%	Steam/cooling/power — pinch can cut 20-30%
Labour + supervision	5-15%	Headcount × shift factor 4.5
Maintenance	3-6% of FCI	Wear-driven

Because raw materials dominate, a 1% yield gain often beats any utility optimisation — test it against a pinch retrofit in the same case.

Profitability. Three lenses, never one:

- *Payback period* (FCI / annual after-tax cash flow): attractive <3 yr, marginal 3-5 yr, suspect >5-6 yr. Ignores time value, so it is a screen only.
- *NPV* at a hurdle rate (10-15% typical): positive ⇒ value-adding; verify sensitivity to discount rate and a 10-year horizon.
- *IRR/DCFROR*: the rate where NPV=0; want it 5-10 points above WACC. Below the hurdle even a short payback fails.

Estimate class governs everything. Report the band, not a point:

AACE class	Maturity	Accuracy	Method
Class 5	0-2%	−30/+50%	Capacity-factored, six-tenths
Class 4	1-15%	−20/+30%	Equipment-factored (Lang)
Class 3	10-40%	−15/+20%	Bare-module / budget
Class 2	30-70%	−10/+15%	Detailed, semi-quantitative

AACE class	Maturity	Accuracy	Method
Class 1	50-100%	−5/+10%	Definitive, quote-based

Quoting an NPV to four significant figures off a Class-5 estimate is the classic glass-box failure: the ±50% CAPEX band swamps the answer. Always run the case at both error bounds. In Choupo the economics pass does it for you: a `sensitivity` block inside the postDict's `economics` block moves each declared input *alone* to two ends and prices the same converged run again — a *tornado*, sorted by NPV swing. `FCI { relative accuracyBand; }` takes the capital to the two ends of the declared band; `price.<stream> { relative ( -0.2 0.2 ); }`, `utility.<name>`, `discountRate`, `streamFactor`, `taxRate` and `labourRate` move the rest (`relative` ends are fractions of the base value, `absolute` ends are values of the input). Read it as a *sensitivity*, not an uncertainty analysis: no probability is attached to either end, and inputs that move together in the market (the hydrogen and the electricity price) are moved one at a time. See `greenAmmonia02_feasibility150bar`, where a ±20% move of the hydrogen or the ammonia price swings the NPV some fourteen times as far as the whole $-30/+50\%$ capital band.

25 Materials of construction & mechanical design

Material choice is a *climb-the-ladder* decision: start with the cheapest metal that survives the duty, and step up only when corrosion rate, chlorides, temperature, or hydrogen force you. Each rung is roughly 1.5–4× the cost of the one below, so an unjustified jump from carbon steel to SS316 can double a vessel's bare cost (see Choupo's Guthrie/Turton material-factor `f_M` in the costing pass; carbonSteel/SS304/SS316/aluminium are the reference `data/materials/*.dat`).

The selection ladder. The governing rule of thumb is a target corrosion rate **< 0.1 mm/yr** (≈ 5 mil/yr) for a 20-year life; 0.1–0.5 mm/yr is tolerable with a fat allowance; above 0.5 mm/yr step up.

Service / limit	Pick	Why (the WHY)
Dry hydrocarbons, steam, mild T < 400 °C	Carbon steel	Cheapest; ~0.1 mm/yr in benign duty
Dilute acids, oxidising, T < 60 °C, **chloride-free**	SS304	Cr_2O_3 passive film; loses it to Cl^-
Chlorides, dilute H_2SO_4, food/pharma	SS316 (2–3% Mo)	Mo resists pitting; PREN ≈ 24 vs 18
Hot chloride + tensile stress	Duplex 2205 / Ni alloy	**Austenitic SCC** kills 304/316
Strong HCl, wet Cl_2, HF	Hastelloy C-276 / lined	General attack defeats stainless
High-purity water, weight-critical	Aluminium	Cheap, light, but soft & low-T only

Chlorides & stress-corrosion (the silent killer). Austenitic 304/316 suffer chloride **SCC** above ≈ **60 °C** with tensile stress and even ppm-level Cl^- — failures look like brittle cracking with no wall loss, so a corrosion allowance does *not* protect you. Below 60 °C or for hot chlorides, jump to duplex (ferritic-austenitic, immune) or nickel alloys. Pitting resistance scales with $\mathbf{PREN = \%Cr + 3.3 \cdot \%Mo + 16 \cdot \%N}$ (304 ≈ 18, 316 ≈ 24, 2205 ≈ 35).

High-T creep & hydrogen. Carbon steel is creep-limited to ≈ **425 °C**; above this design stress collapses, so 1¼Cr–½Mo, 9Cr, or austenitics take over (relevant for fired-heater tubes and HiTec-salt service to 535 °C). In H_2 at pressure, consult **Nelson curves** (API 941): below the curve carbon steel is safe; above it, hydrogen attacks carbides → decarburisation and fissuring, demanding Cr-Mo steels.

Corrosion allowance. Add **3.0 mm** standard (1.5 mm benign, up to 6 mm aggressive). It is sacrificial metal, *not* an SCC or hydrogen fix.

Wall thickness vs pressure. Thin-wall (ASME VIII Div. 1): $\mathbf{t = P \cdot R/(S \cdot E - 0.6 \cdot P) + CA}$, with weld efficiency E = 0.85–1.0 and S the allowable stress (≈ 138 MPa for SS316 at moderate T). Design pressure margin: **max(operating + 10%, +1.7 bar)**; design temperature: operating **+ 25 °C**. A minimum shell of ~6–8 mm is set by handling, not pressure.

Aspect ratios. Vertical process vessels and knockout drums: **L/D ≈ 3** (range 2–5) — minimises wall metal at fixed volume while giving disengaging height. Distillation columns run **L/D up to 20–30**, capped by wind/seismic moment. Atmospheric storage tanks favour **L/D ≈ 0.5–1** (squat, cheap floor). Horizontal drums sit at **L/D ≈ 3–4**.

Glass-box check: re-run the Choupo sizing/costing pass swapping `material carbonSteel;` → `SS316;` and watch `f_M` and shell mass move; that is the cost of one rung.

Corrosion allowance — how many millimetres? Wall thickness = the pressure-stress thickness (from ASME) **plus a sacrificial corrosion allowance**: CA = corrosion rate × design life. Carbon steel corroding at ~0.1–0.25 mm/yr over a 20–25 year life gives the familiar **3 mm** (1/8 in) allowance; severe service runs to **6 mm**, clean/inert service to **0–1.5 mm**, and a corrosion-resistant alloy needs **none** (it is chosen so the rate is ~0).

Corrosion rate	Verdict
< 0.1 mm/yr	excellent — carbon steel, 1.5–3 mm allowance
0.1–0.5 mm/yr	acceptable with allowance + inspection
> 0.5 mm/yr	do NOT just thicken the steel — upgrade the alloy or line it

The rule: a corrosion allowance buys time, not safety — above ~0.5 mm/yr the economics flip to a better material.

Gaskets — the small part that floods the floor. A gasket seals a flange by being squeezed between the faces; it must survive the **pressure, the temperature, AND every fluid the joint ever sees — including cleaning.** Select by P–T–fluid:

Gasket	P–T window	Use when
Compressed fibre / PTFE sheet	to ~20 bar, ~200 °C	low duty; PTFE for chemical resistance
Spiral-wound (metal + graphite/PTFE filler)	to ~100+ bar, ~450 °C+	the workhorse for process flanges
Ring-joint (RTJ, solid metal)	very high P (> 100 bar)	high-pressure gas, wellheads

The wetted material — the winding and the filler — must resist the worst fluid, and the worst fluid is often **not** the product but the **CLEANING** step. A gasket chosen only for cold wine will corrode through during a **40 °C acid CIP wash**, and then the wine runs out of the joint on the next batch — a real and classic failure. **Specify the gasket (and the bolt load) for the most aggressive condition the joint ever sees, cleaning included**, never for normal operation alone.

26 Standards & further reading

This guide distils the classic references below. For any real design go to the primary source — and for anything safety- or environment-critical, the governing standard is the authority, not this guide (see *Disclaimer & scope*).

General process design & heuristics

- Turton, Shaeiwitz, Bhattacharyya & Whiting, *Analysis, Synthesis, and Design of Chemical Processes*, Prentice Hall.
- Seider, Seader, Lewin & Widagdo, *Product and Process Design Principles*, Wiley.
- Sinnott & Towler, *Chemical Engineering Design* (Coulson & Richardson, Vol. 6), Butterworth-Heinemann.
- Green & Perry, *Perry's Chemical Engineers' Handbook*, McGraw-Hill.
- Douglas, *Conceptual Design of Chemical Processes*, McGraw-Hill.
- Branan, *Rules of Thumb for Chemical Engineers*; Walas, *Chemical Process Equipment.*

By topic

- Heat integration / pinch — Smith, *Chemical Process Design and Integration*, Wiley; Kemp, *Pinch Analysis and Process Integration.*
- Distillation — Kister, *Distillation Design & Distillation Operation*, McGraw-Hill.
- Heat transfer — Kern, *Process Heat Transfer*; **TEMA** standards.
- Reactors — Fogler, *Elements of Chemical Reaction Engineering*; Levenspiel, *Chemical Reaction Engineering.*
- Membranes — Baker, *Membrane Technology and Applications*, Wiley.
- Plant layout — Mecklenburgh (ed.), *Process Plant Layout*, IChemE; Bausbacher & Hunt, *Process Plant Layout and Piping Design*, Prentice Hall; CCPS, *Guidelines for Siting and Layout of Facilities*, AIChE.
- Materials / corrosion — **NACE/AMPP** standards; **API RP 571** (damage mechanisms).
- Effluent — Metcalf & Eddy, *Wastewater Engineering*; **US EPA** and **EU Industrial Emissions Directive (BAT / BREF)** guidance.

Safety & environmental standards — binding, not optional

- Pressure relief & venting — **API STD 520 / 521 / 526 / 2000**; **ASME BPVC Sec. VIII & I**.
- Flares — **API STD 521 / 537**.
- Process-safety management — **OSHA 29 CFR 1910.119**; **AIChE CCPS** guideline series.
- Functional safety / SIS — **IEC 61508 / 61511**; HAZOP — **IEC 61882**.
- Flammables — **NFPA 30**; electrical area classification — **NFPA 70 / IEC 60079**.
- Siting & spacing — **API RP 752 / 753** (occupied buildings); **NFPA 30 / 58**, **API STD 2510** (tank and LPG spacing); hazardous areas — **IEC 60079-10-1**, **EI Model Code Part 15**, **API RP 505**, **NFPA 497**, **ATEX 1999/92/EC**.
- Emissions & discharge — the local regulator's limits (e.g. **US EPA**, **EU IED / BAT-BREF**).

27 Equipment selection at a glance

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28 Equipment selection at a glance

This one-page table is the *index of bets* — each row is a duty you will face, the default pick, the one-line reason, and the range to start from. Treat every "typical number" as a hypothesis to verify by running the linked Choupo case. When two picks compete, the table gives the **default**; the section explains the crossover.

Duty / situation	Pick this	Because	Typical numbers
Overall flowsheet structure	Reactor $\rightarrow$ separation $\rightarrow$ recycle, with a purge on any loop that accumulates inerts	The "onion": fix conversion + recycle before heat/utilities; purge bleeds what recycle would otherwise trap	Purge loses 1–5 % of recycle; target recycle so single-pass conversion $\neq$ overall conversion
Equilibrium-limited reaction	Push reactor to ~90–95 % of equilibrium, recycle the rest	Last few % of approach costs disproportionate volume; recycle is cheaper than oversizing	Single-pass $X = 0.6$–0.9; overall $X > 0.99$ via recycle
Fast liquid-phase reaction, high conversion	PFR (or CSTR-in-series)	Plug flow needs least volume for a given conversion; back-mixing penalises high X	PFR vol $\approx$ 1/3–1/2 of one CSTR at $X = 0.9$
Autocatalytic, biological, solids-laden, or strong heat-removal duty	CSTR (or CSTR train)	Back-mixing dilutes feed, eases T control and solids handling	3–4 CSTRs in series $\approx$ PFR; τ set by k and target X
Gas-phase catalytic reaction	Packed-bed (PFR model)	Heterogeneous catalyst + plug flow; watch ΔP across bed	Superficial gas 0.3–1 m/s; bed ΔP a few % of P
Separate close-boiling liquids	Distillation	Cheapest workhorse whenever relative volatility allows	Viable for $\alpha >$ ~1.05; easy for $\alpha > 1.5$
Distillation column operating point	Reflux R = 1.1–1.3 × R_min	Optimum trades reflux (energy) vs trays (capital); flat-bottomed cost curve	R/R_min $\approx$ 1.1–1.3; N $\approx$ 2–2.5 × N_min
Heat-sensitive / high-boiling bottoms	Vacuum distillation	Lowers boiling T below decomposition; protects product	Run at 0.05–0.3 bar when T_bottom > ~150–200 °C at 1 atm
Column vapour loading	Size to F-factor / fraction of flooding	Avoids flooding (carry-over) and weeping	F = u · √ρ_v $\approx$ 1–2.5; design at 70–85 % of flood
Dilute aqueous dewatering / desalination	Reverse osmosis (spiral-wound)	Pressure-driven, no phase change; rejects monovalent salts	ΔP 15–70 bar (seawater); flux 15–30 LMH
Recover solvent / reject divalent ions, pass monovalent	Nanofiltration	Looser membrane, lower pressure than RO, partial-rejection selectivity	ΔP 5–20 bar; rejects multivalent > 90 %, monovalent 20–70 %
Remove particles from a gas	Cyclone (coarse) $\rightarrow$ bag filter (fine)	Cyclone is cheap above ~5–10 µm; fabric filter takes sub-micron	Cyclone η high > 5–10 µm; bag filter to < 1 µm
Produce a crystalline solid	MSMPR cooling crystalliser (or evaporative if solubility flat with T)	Cooling exploits steep solubility-T slope; evaporative when it is flat	Supersaturation low (metastable); residence min–hours
Clean liquid–liquid heat transfer	Shell-and-tube (or gasketed plate if T,P moderate)	Robust, costed, high U; plates beat it on area/cost when clean & < ~25 bar	U $\approx$ 250–750 W/m^{2}K; ΔT_min $\approx$ 10 °C
Heat transfer with a gas on one/both sides	Extended/compact surface — **not** a bare shell-and-tube	Gas-side h is the bottleneck; gas–gas shell-and-tube needs huge area	U_gas–gas $\approx$ 10–50 W/m^{2}K; ΔT_min $\approx$ 20 °C

Duty / situation	Pick this	Because	Typical numbers
Condense overhead / reboil bottoms	Kettle/thermosiphon reboiler; shell-side condenser	Phase change gives very high h; match utility to T-level	U_cond ≈ 1000–3000 W/m^2K; ΔT_min ≈ 5–10 °C
Process duty above ~400 °C	Fired heater (very high T) or molten-salt loop (250–535 °C)	Steam tops out ~250 °C; salt covers the gap, fired heater above	HiTec salt 142–535 °C; Dowtherm A to ~400 °C
Size a pumped liquid line	Velocity 1–3 m/s (suction lower)	Balances pumping ΔP vs pipe capital; avoids settling/erosion	Discharge 1–3 m/s; suction 0.5–1.5 m/s; ΔP ~0.5 bar/100 m
Size a gas/vapour line	Velocity 15–30 m/s, capped by ρu^2	Low-density fluid → high velocity OK; erosion-corrosion limits ρu^2	15–30 m/s; steam 30–60 m/s; ρu^2 < ~10000 kg/m · s^2
Default material of construction	Carbon steel	Cheapest; specify up only when corrosion/T demands it	Use to ~425 °C, non-corrosive, no chlorides
Chloride or aqueous-corrosive service	SS316 (not SS304)	Mo content resists chloride pitting that defeats 304	316 for chlorides/acids; 304 for mild aqueous/atmospheric

Use this guide as a map, not a destination. Every range here is a place to *start* a Choupo case, not a number to copy onto a P&ID. The discipline the guide is teaching is the loop itself: form the heuristic bet, build the smallest case that tests it, run it, read what the solver actually does, and let the KPIs confirm or move the range for your system. The rules that survive that loop become judgment; the ones that fail teach you where your problem lives outside the textbook's assumptions. That is the whole point of a glass-box simulator — and the reason a heuristic you have personally falsified once is worth ten you merely memorised. Read the sections in the onion order, keep the at-a-glance table beside the keyboard, and verify everything by running it.